INCH-POUND
MIL-DTL-83507/10A
25 July 2025
SUPERSEDING
MIL-T-83507/10
15 Nov 1977

DETAIL SPECIFICATION

TOOL KITS, ELECTRICAL CONNECTORS, CONTACTS, AND CONNECTOR AND CABLE ACCESSORIES: TOOL KIT FOR ON089559, ON089560, AND ON089561 CONNECTORS

Inactive for new design after September 1, 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the tool kit described herein shall consist of this document and the latest issue of Specification MIL-DTL-83507.

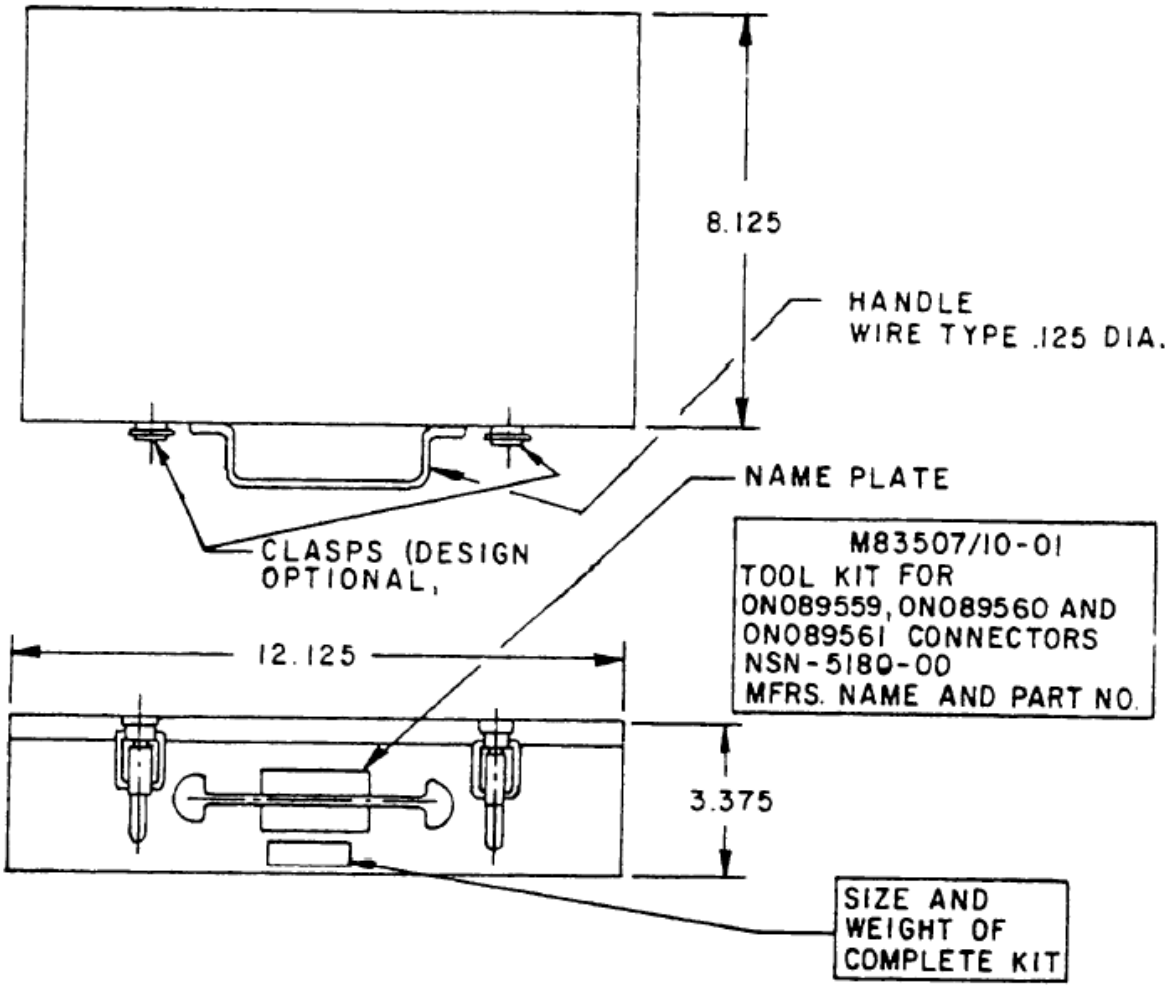


FIGURE 1 Metal case. Tool kit for ON089559, ON089560, and ON089561 connectors.

AMSC N/A FSC 5180

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:
1. Part number M83507/10-01 includes case and tools.
2. Part number M83507 /10-02 for case only (tools not included).

PARTS LIST:

(1)	M83507/10-02	-	Tool case assembly.
(1)	M22520/1-01	-	Crimp tool frame in accordance with SAE-AS22520/1.
(1)	M22520/2-01	-	Crimp tool frame in accordance with SAE-AS22520/2.
(1)	M22520/10-01	-	Crimp tool frame in accordance with SAE-AS22520/10.
(1)	M22520/1-02	-	Turret head in accordance with SAE-AS22520/1.
(1)	M22520/2-24	-	Positioner in accordance with SAE-AS22520/2.
(1)	M22520/10-06	-	Die set in accordance with SAE-AS22520/10.
(1)	M22520/3-1	-	In-service inspection gage in accordance with SAE-AS22520/3.
(1)	9/64 hex wrench.		
(2)	ON089564	-	Installing tool.
(2)	ON089565	-	Removal tool.
(3)	M81969/14-03	-	Installing and removal tool in accordance with SAE-AS81969/14.
(3)	M81969/14-02	-	Installing and removal tool SAE-AS81969/14.

(1 set) Contents charts, instruction sheets, and tool selection chart.

FIGURE 1 Metal case. Tool kit for ON089559, ON089560, and ON089561 connectors. – Continued

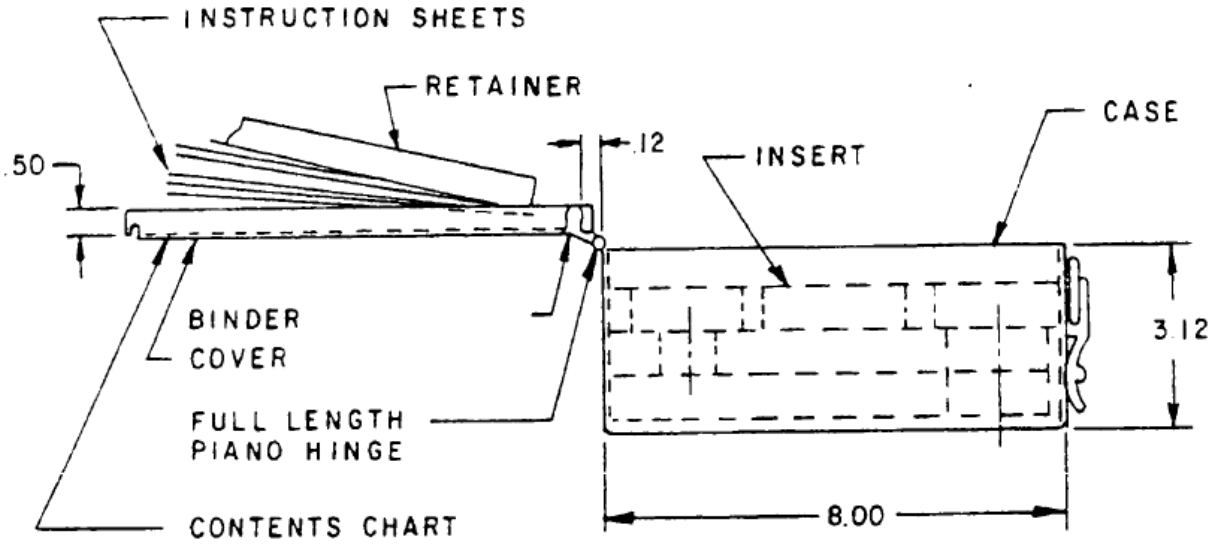


FIGURE 2 Metal case. Tool case assembly..

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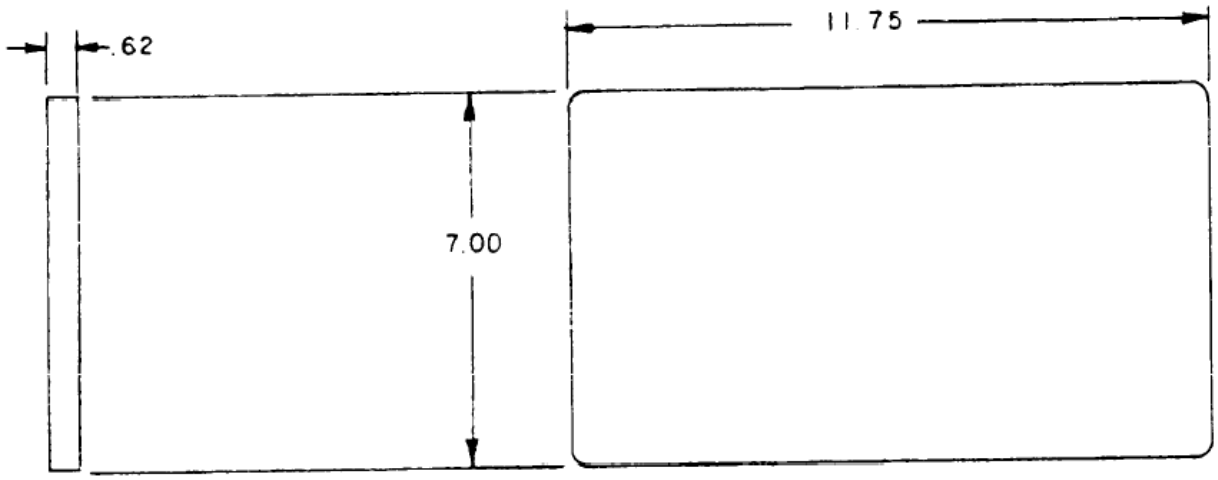


FIGURE 3 Metal case. Retainer

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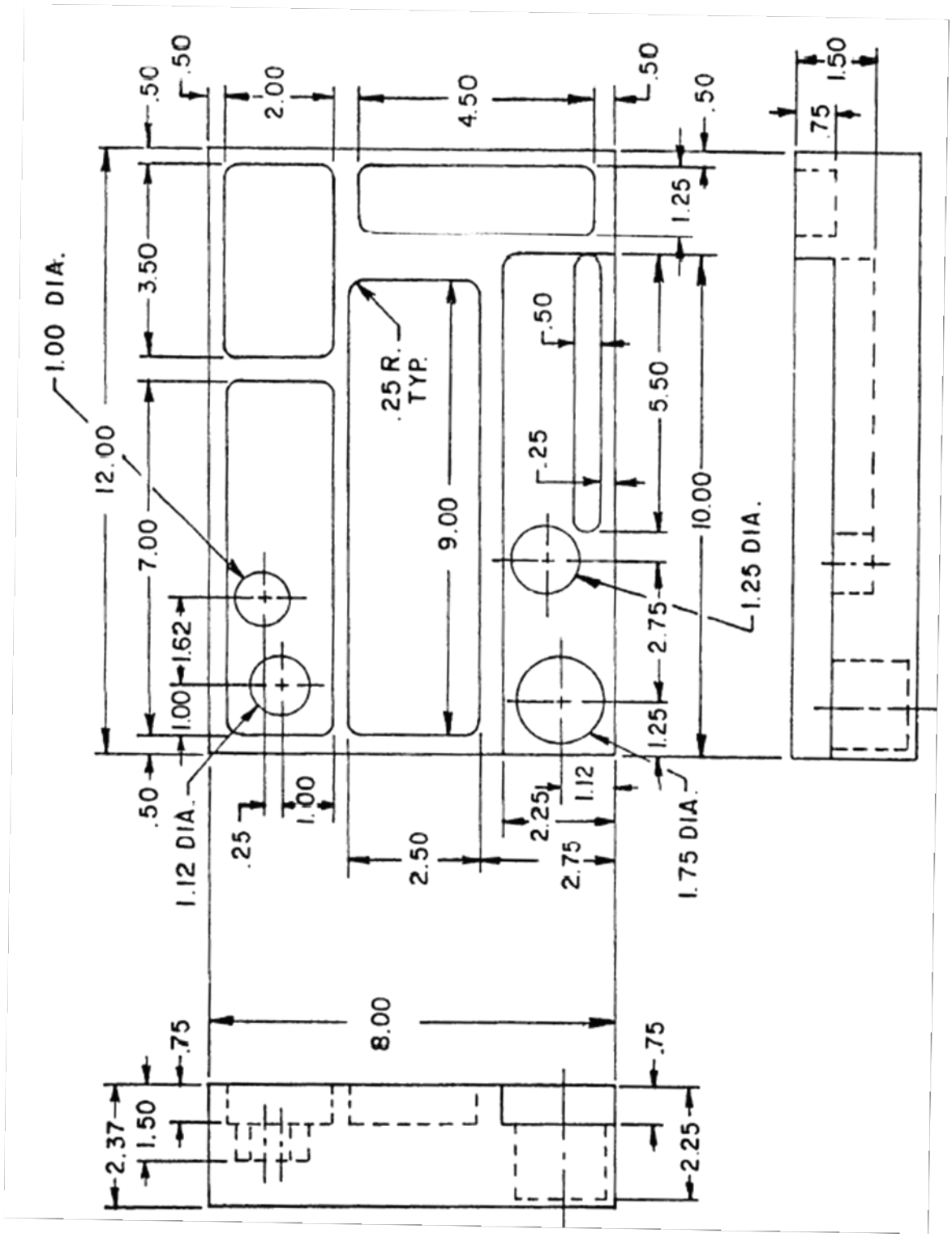


FIGURE 4 Metal case. Insert..

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Instruction sheets are as follows:

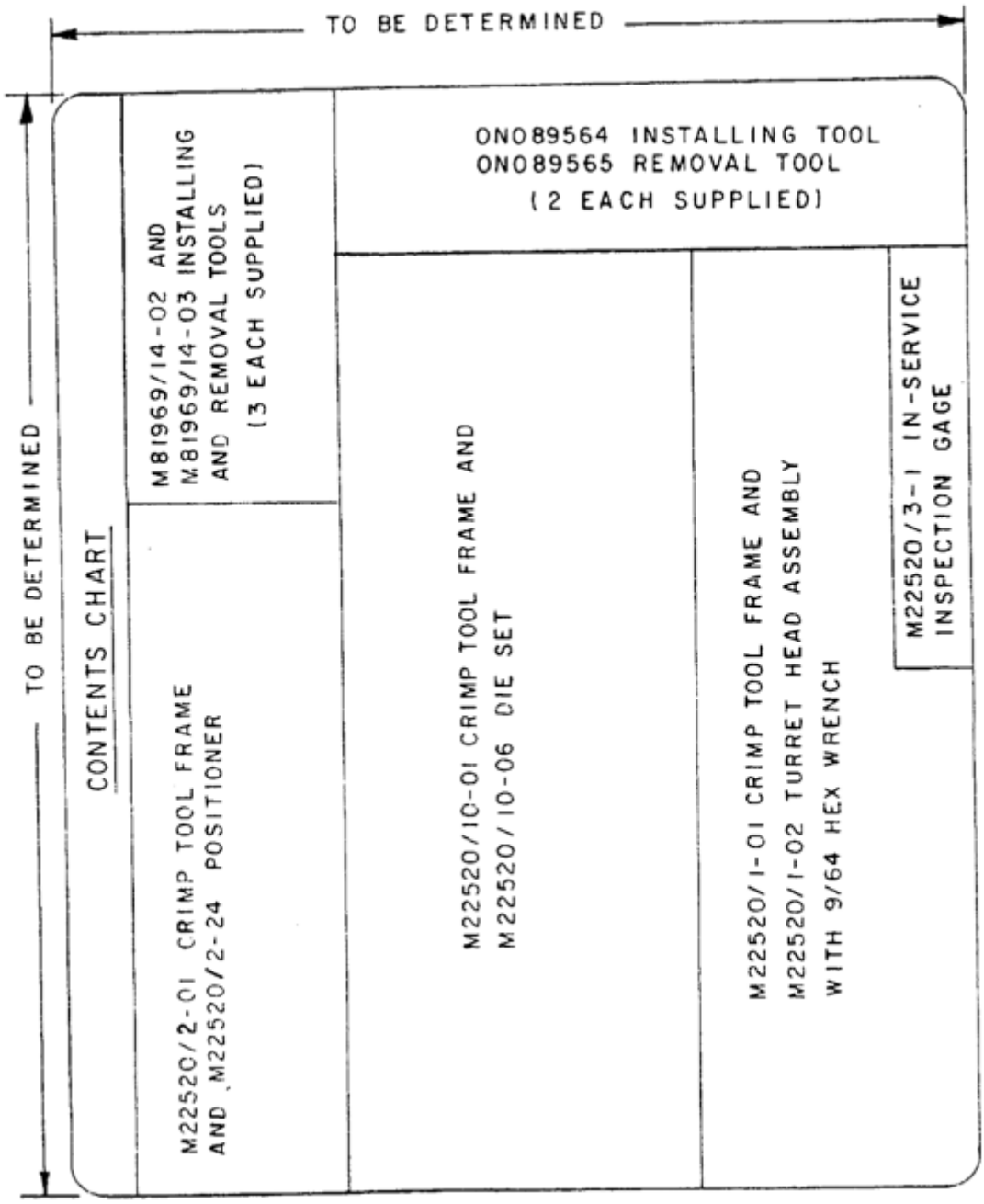


FIGURE 5 Metal case. Instruction sheet.

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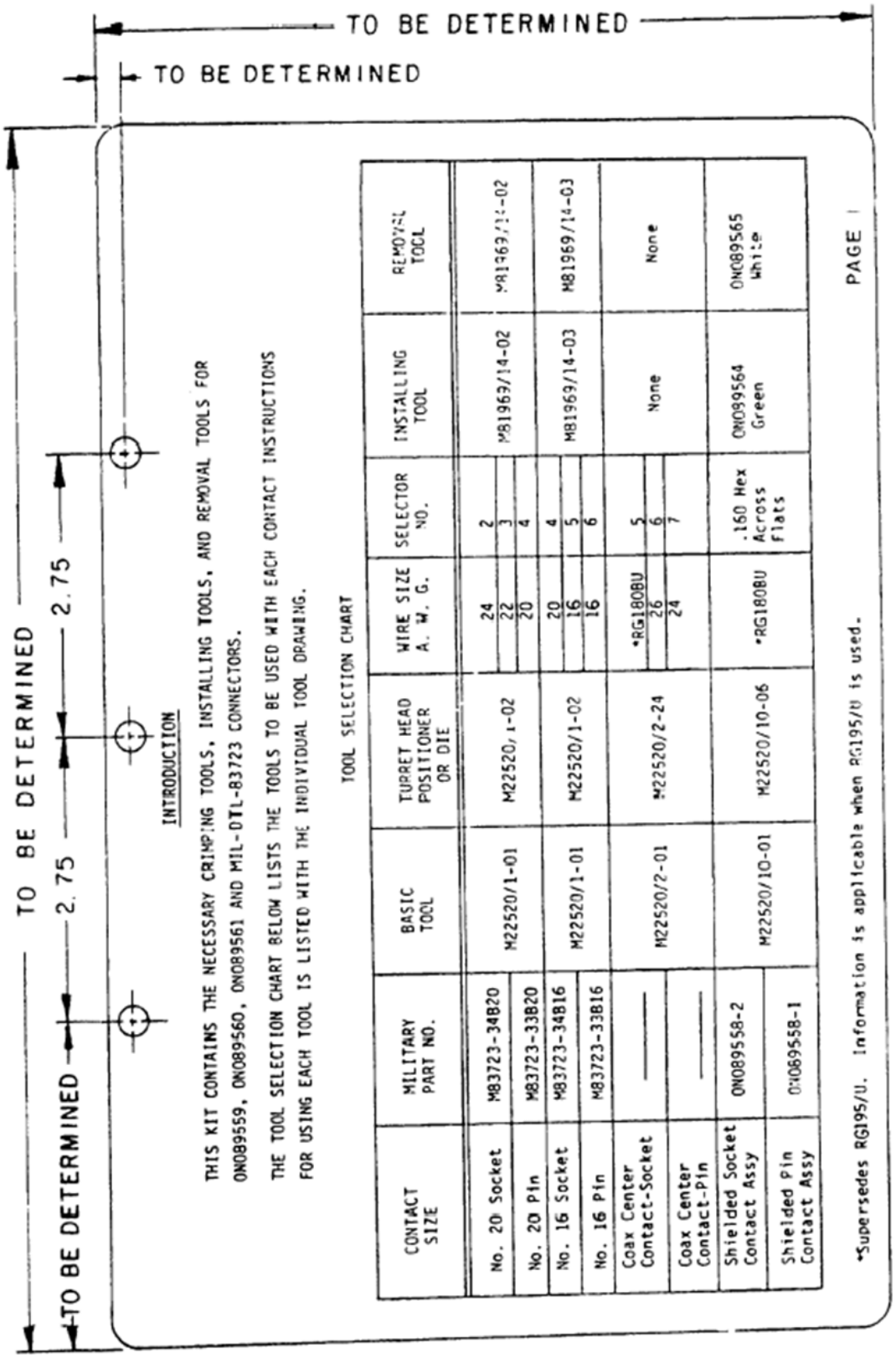


FIGURE 6 Metal case. Instruction sheet.

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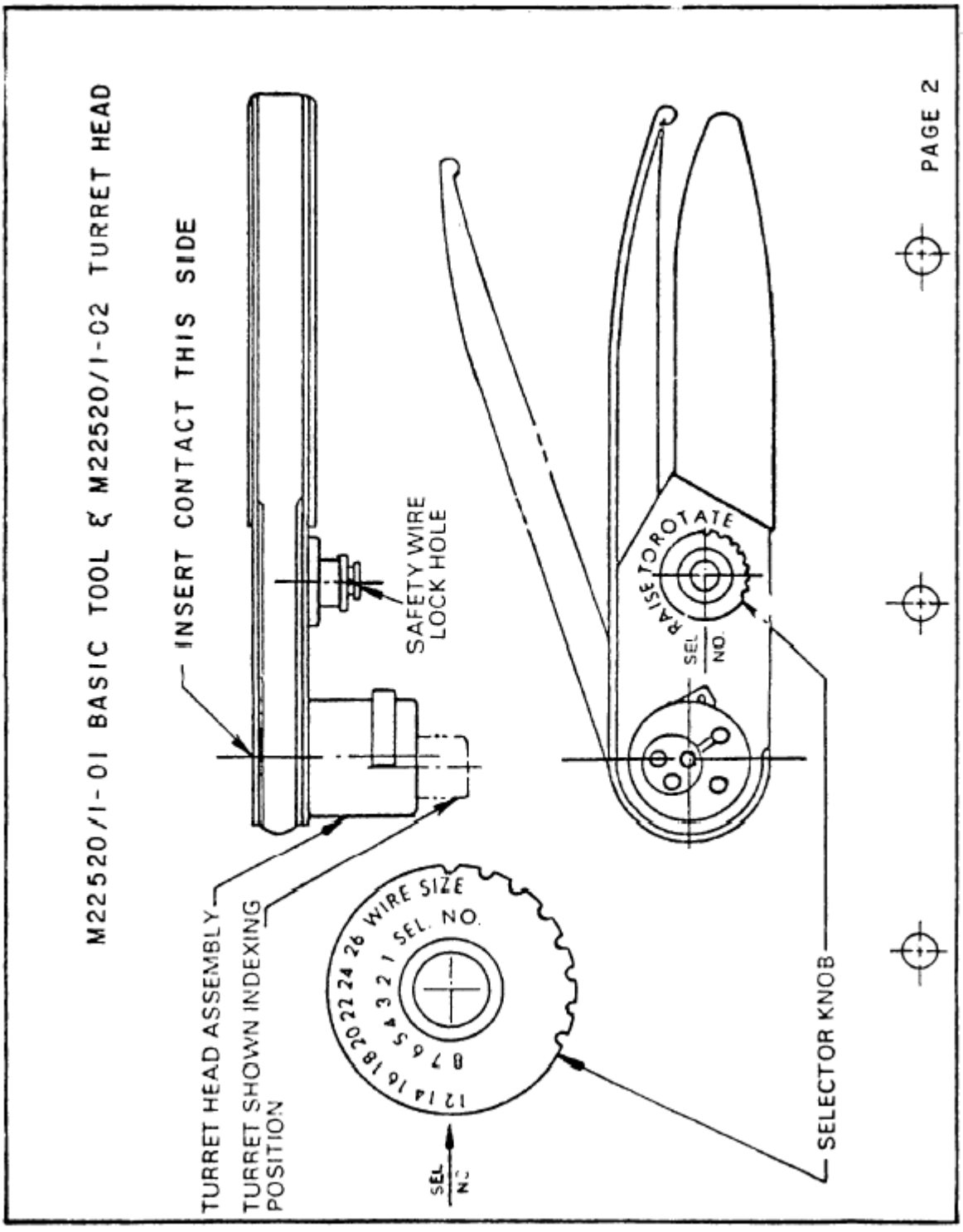


FIGURE 7 Metal case. Instruction sheet.

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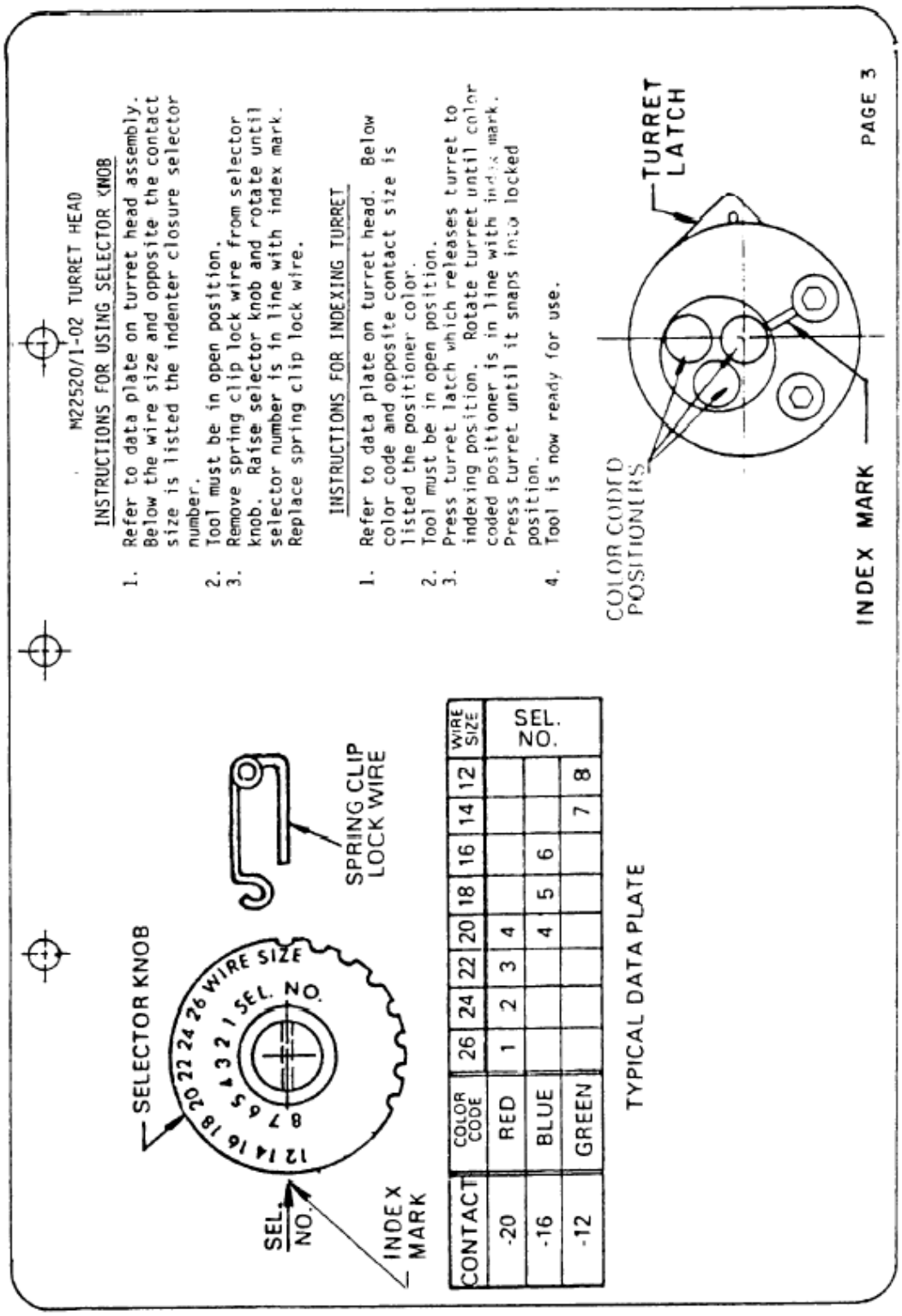


FIGURE 8 Metal case. Instruction sheet.

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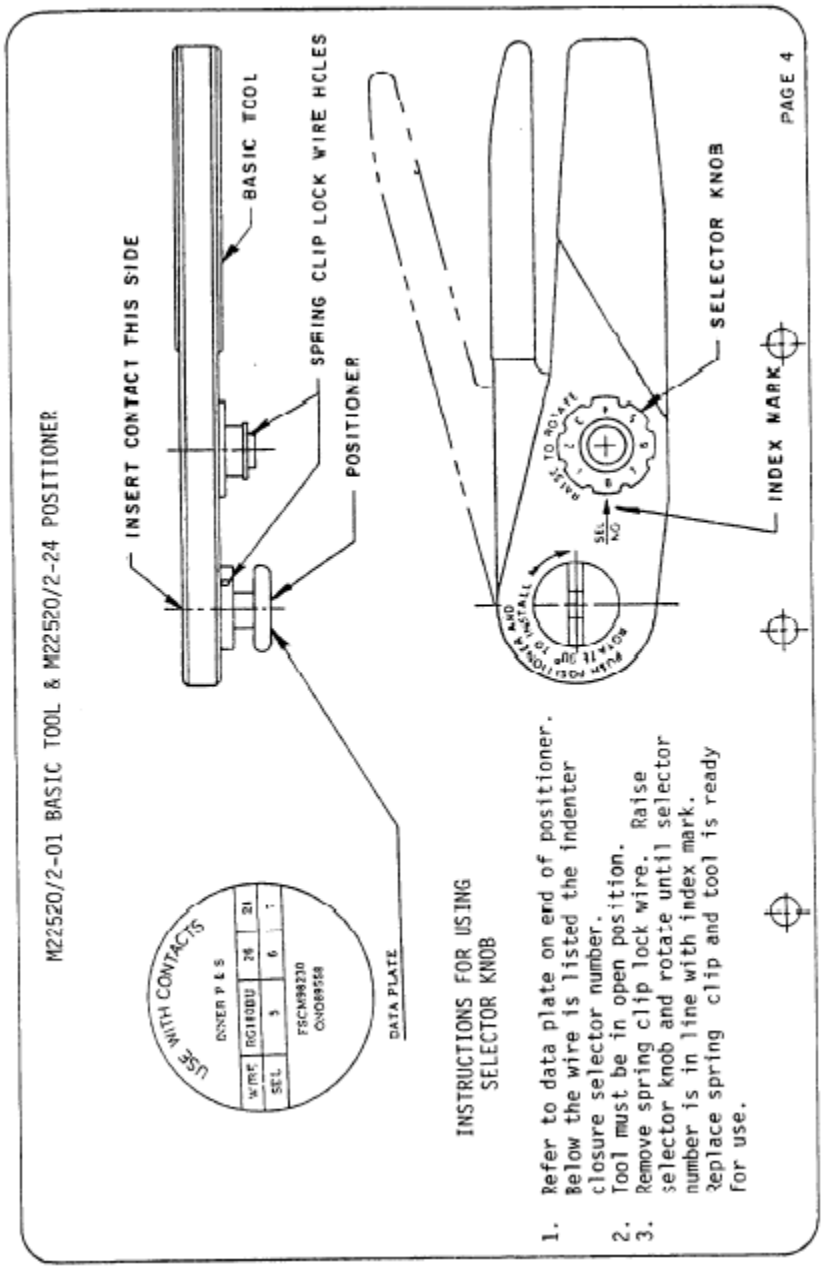


FIGURE 9 Metal case. Instruction sheet.

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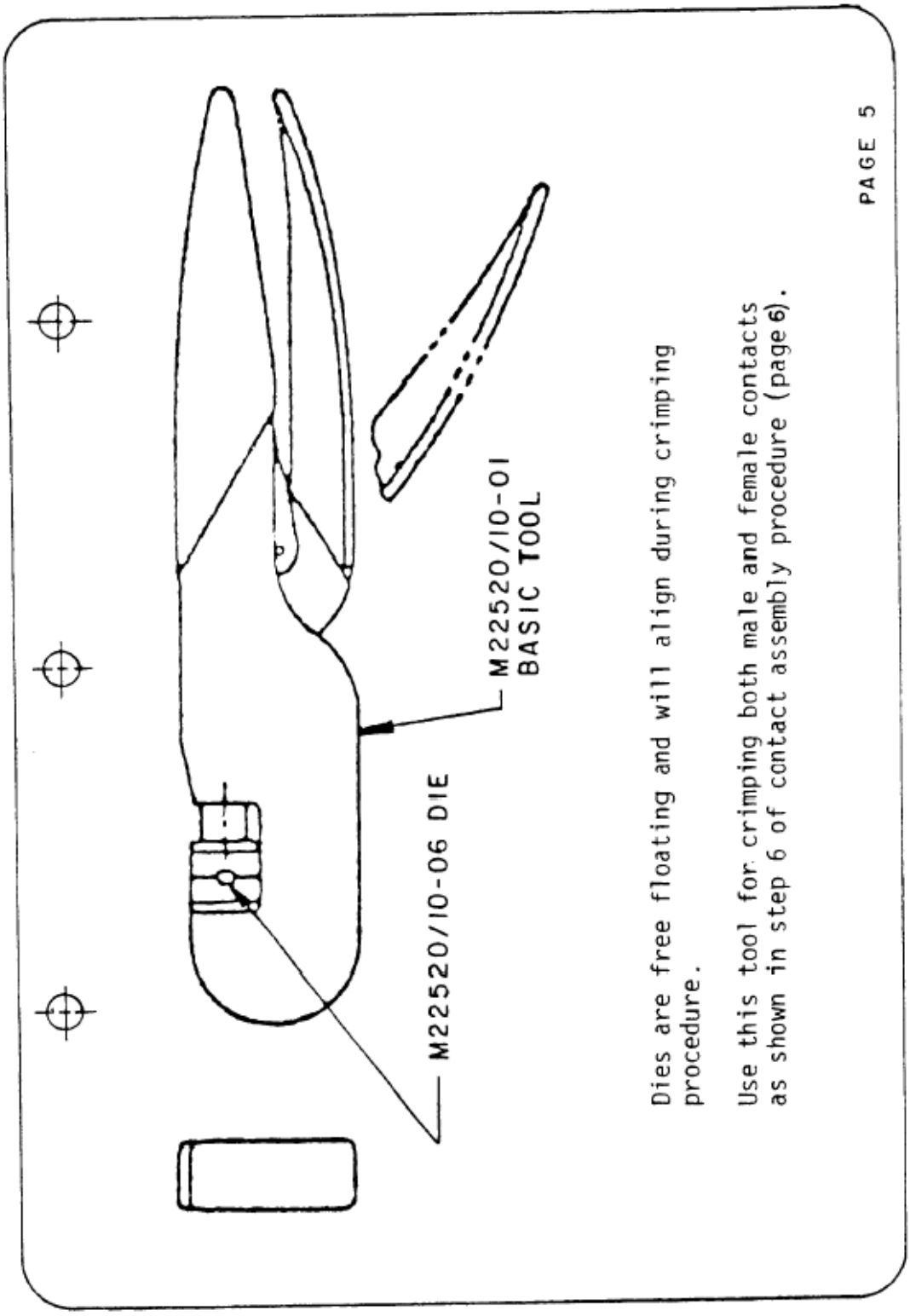


FIGURE 10 Metal case. Instruction sheet.

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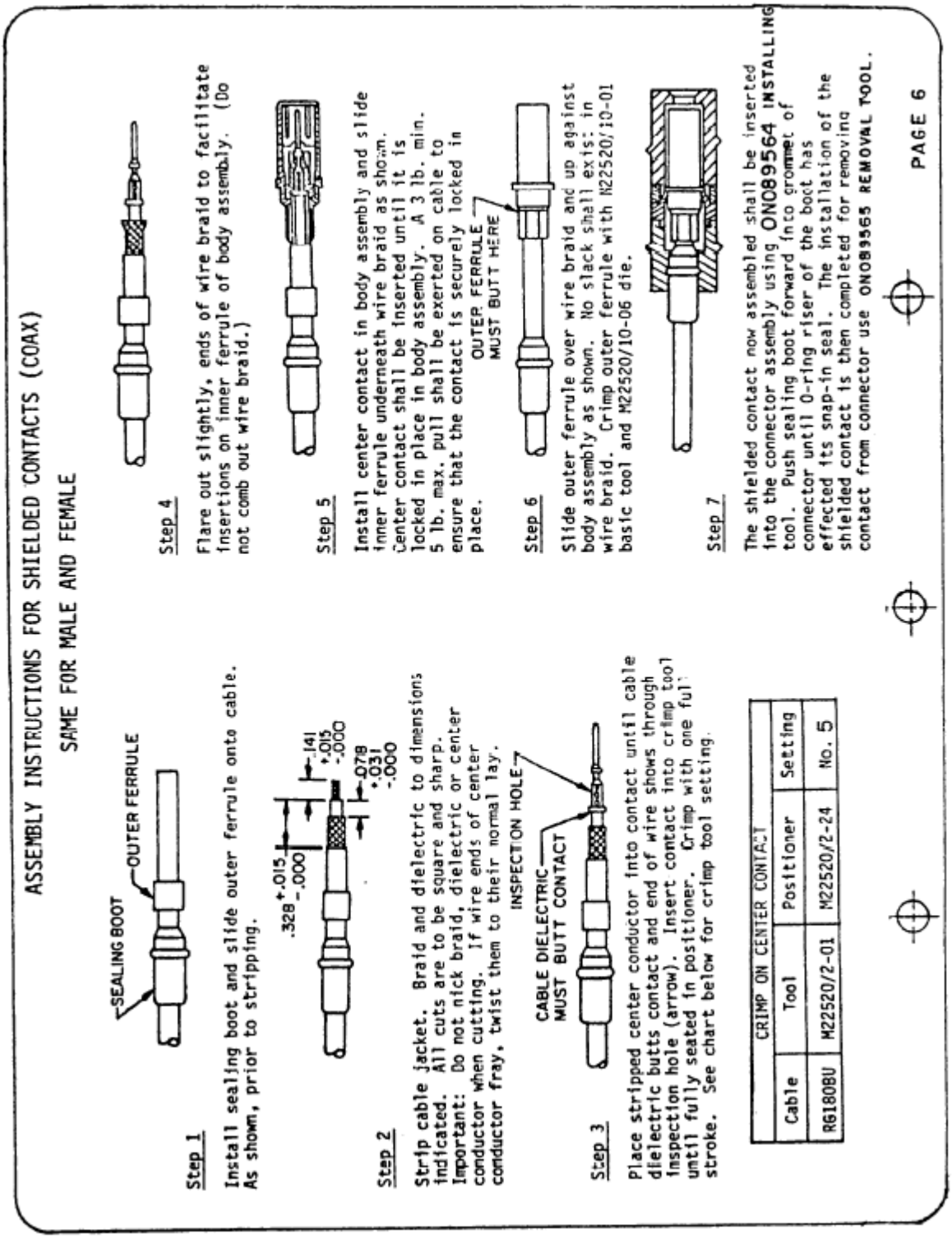


FIGURE 11 Metal case. Instruction sheet.

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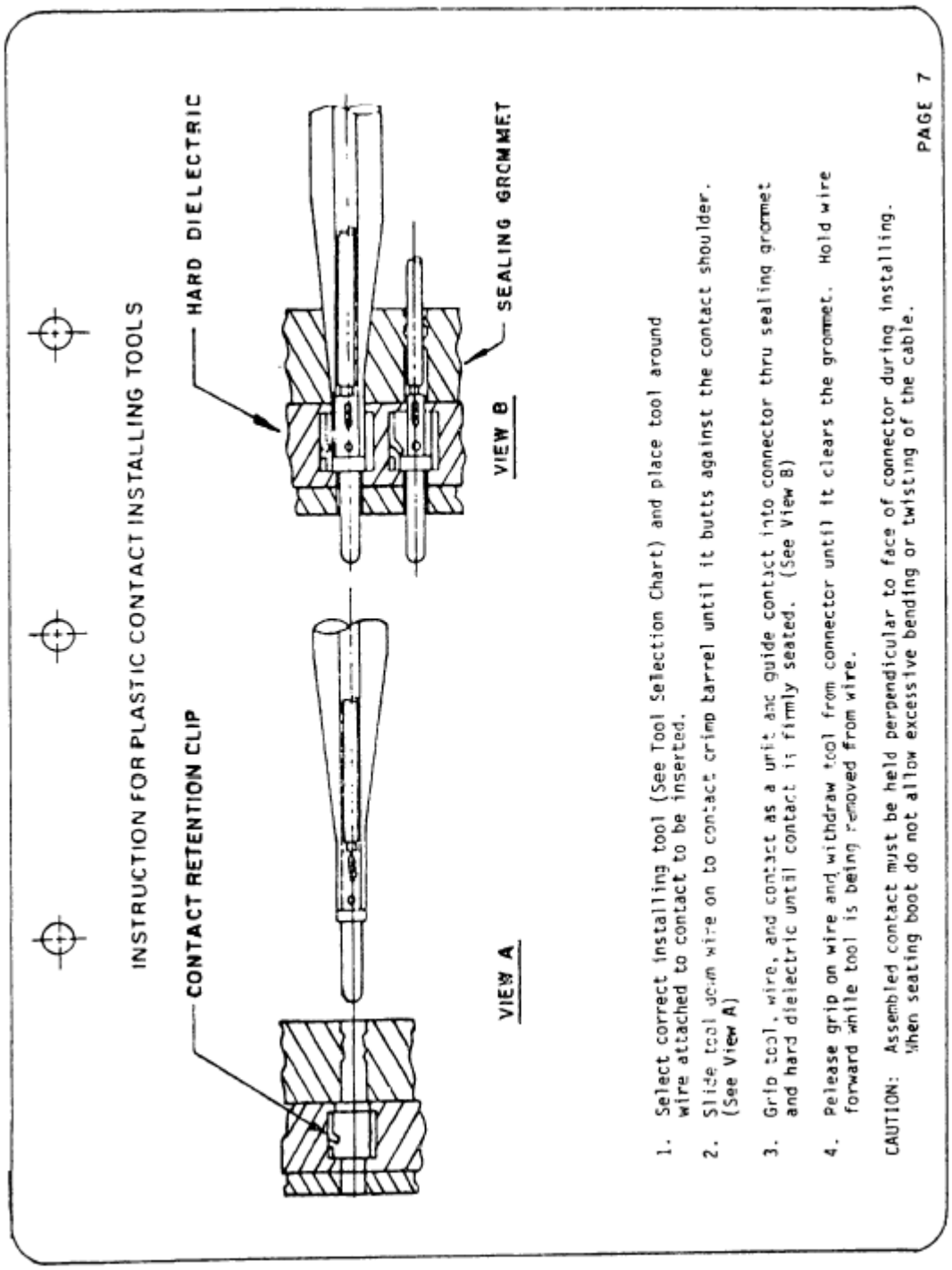


FIGURE 12 Metal case. Instruction sheet.

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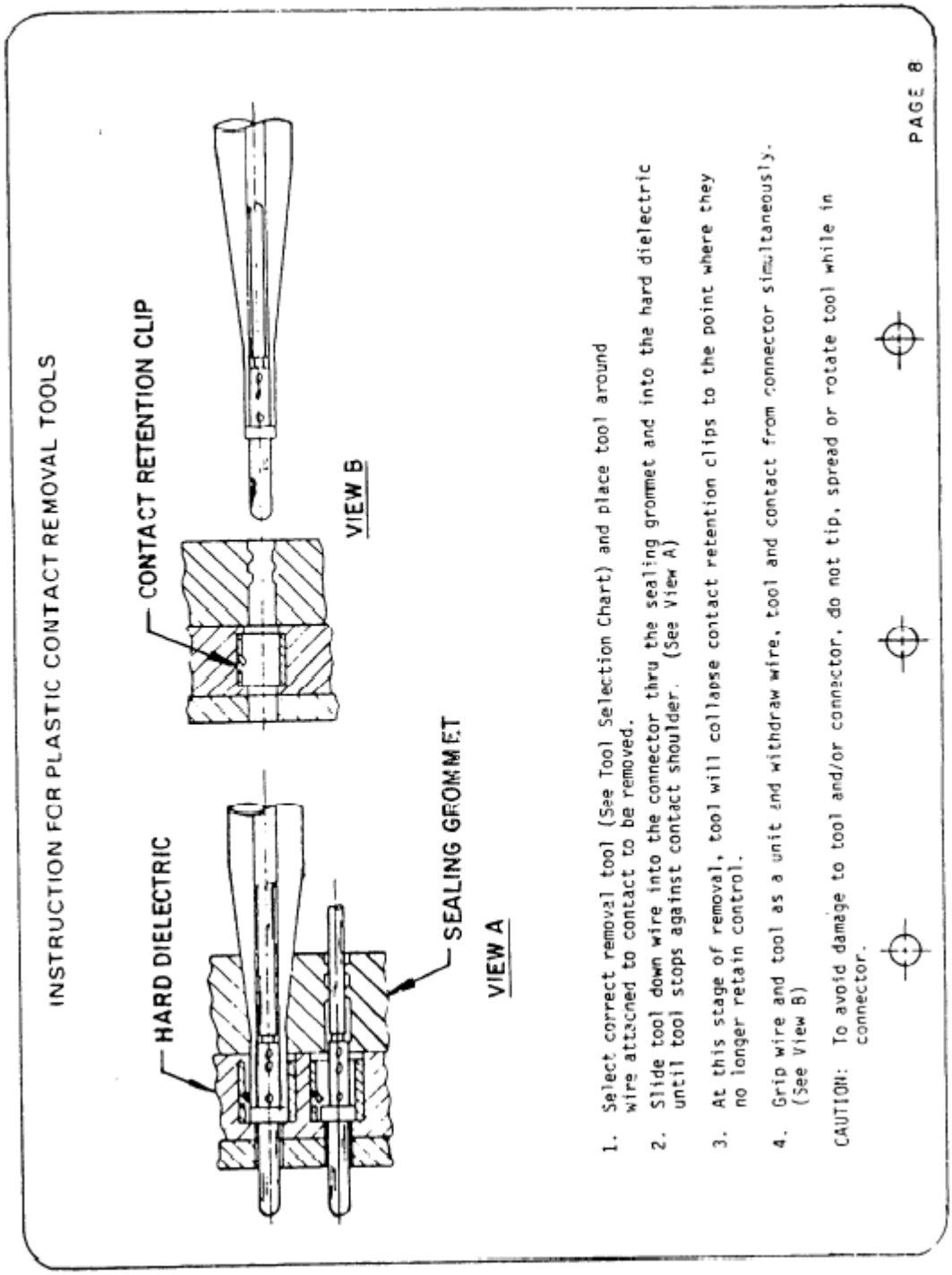


FIGURE 13 Metal case. Instruction sheet.

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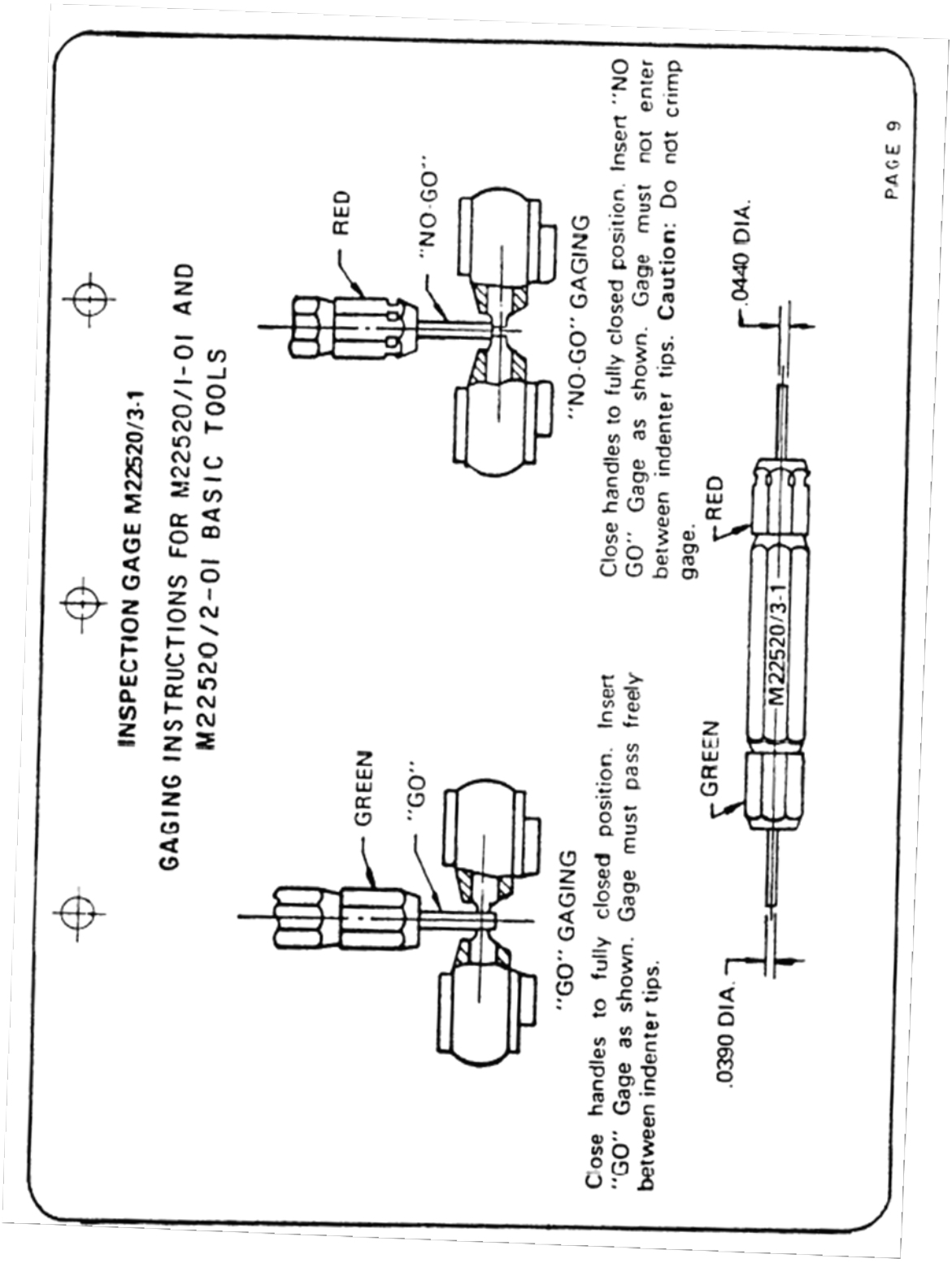


FIGURE 14 Metal case. Instruction sheet.

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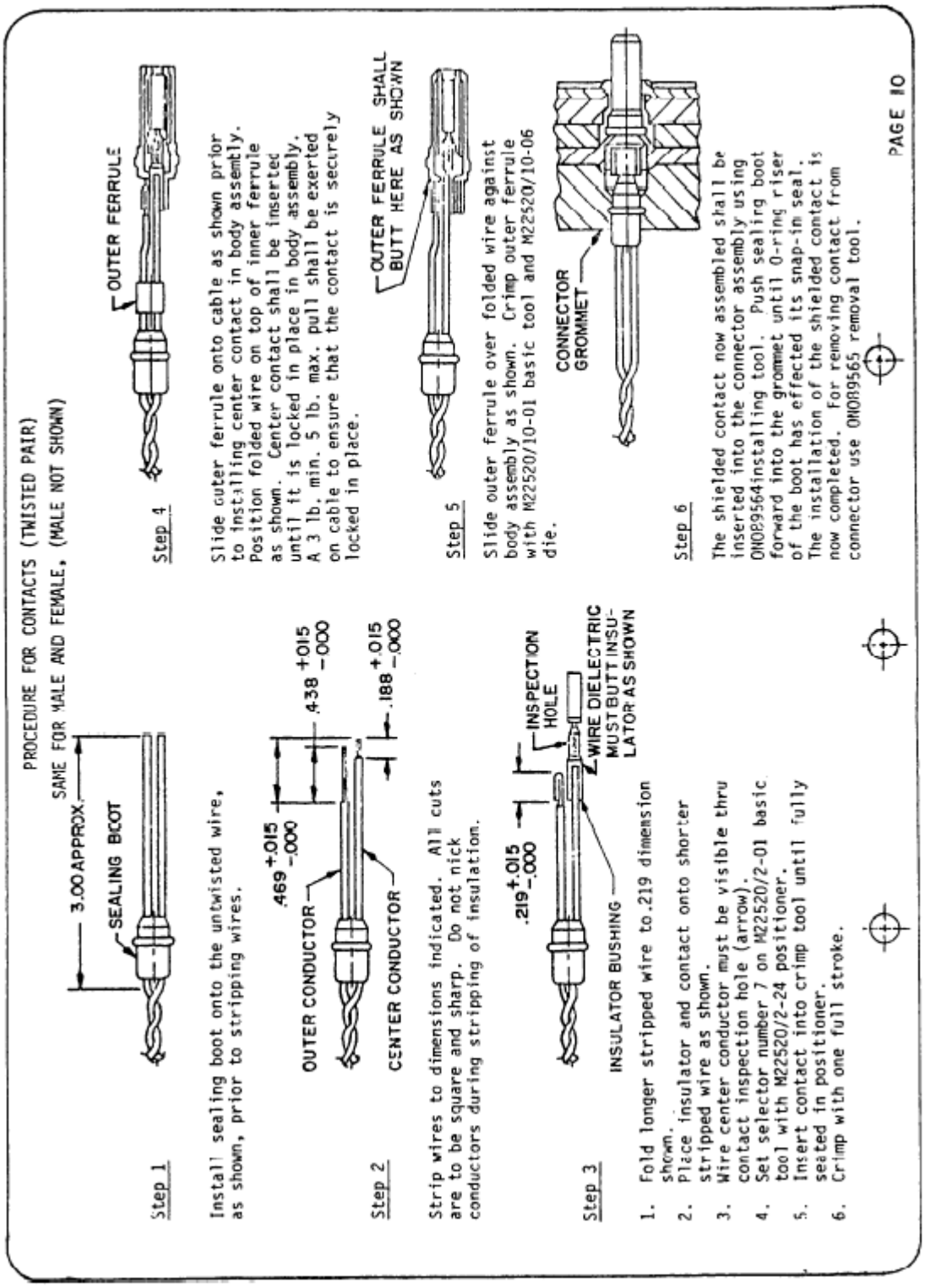


FIGURE 15 Metal case. Instruction sheet.

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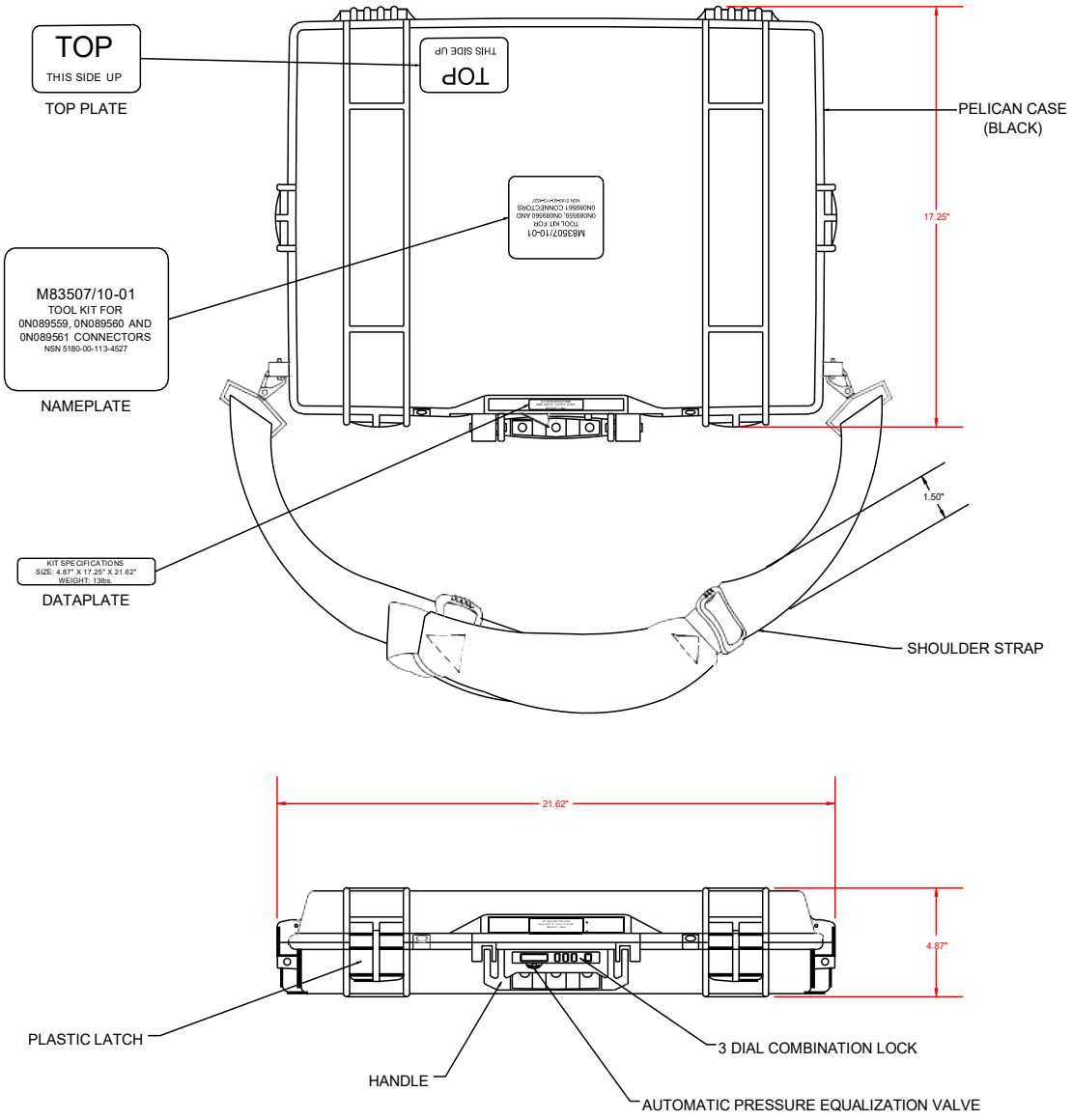


FIGURE 16 Plastic case. TOOL KIT FOR 0N089559, 0N089560 AND 0N089561 CONNECTORS.

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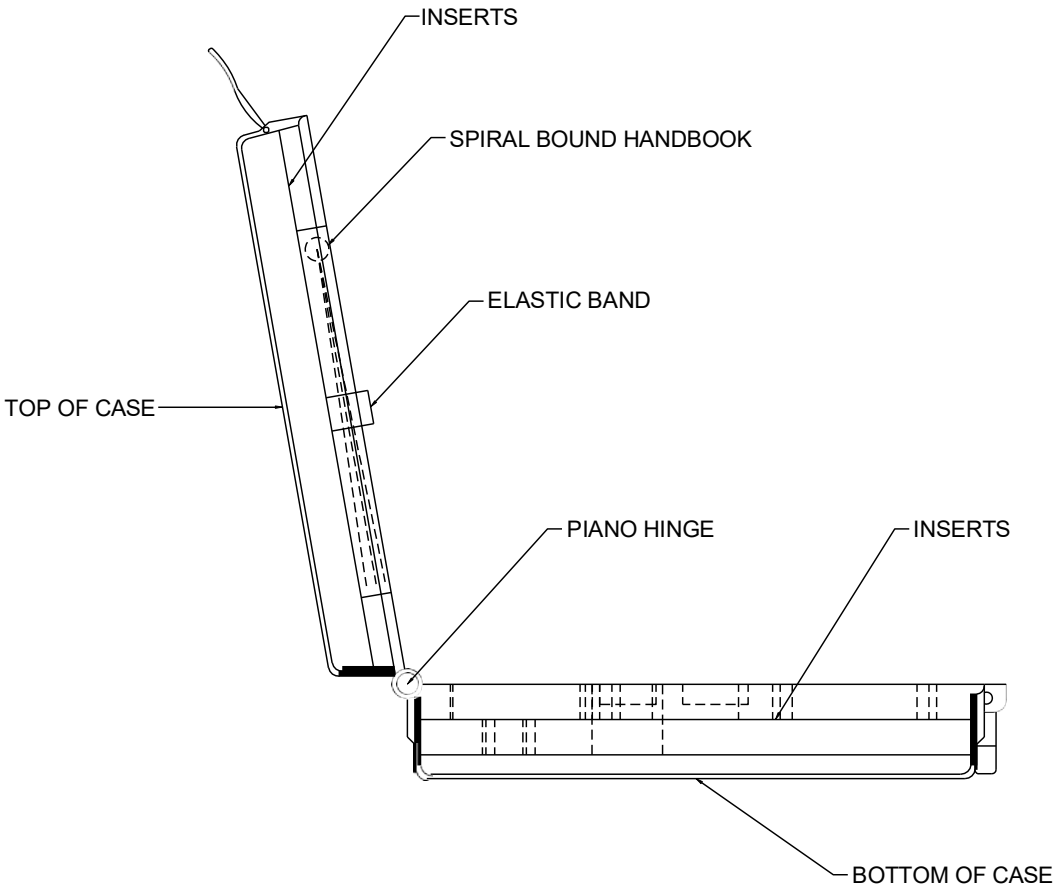


FIGURE 17 Plastic case. TOOL CASE ASSEMBLY.

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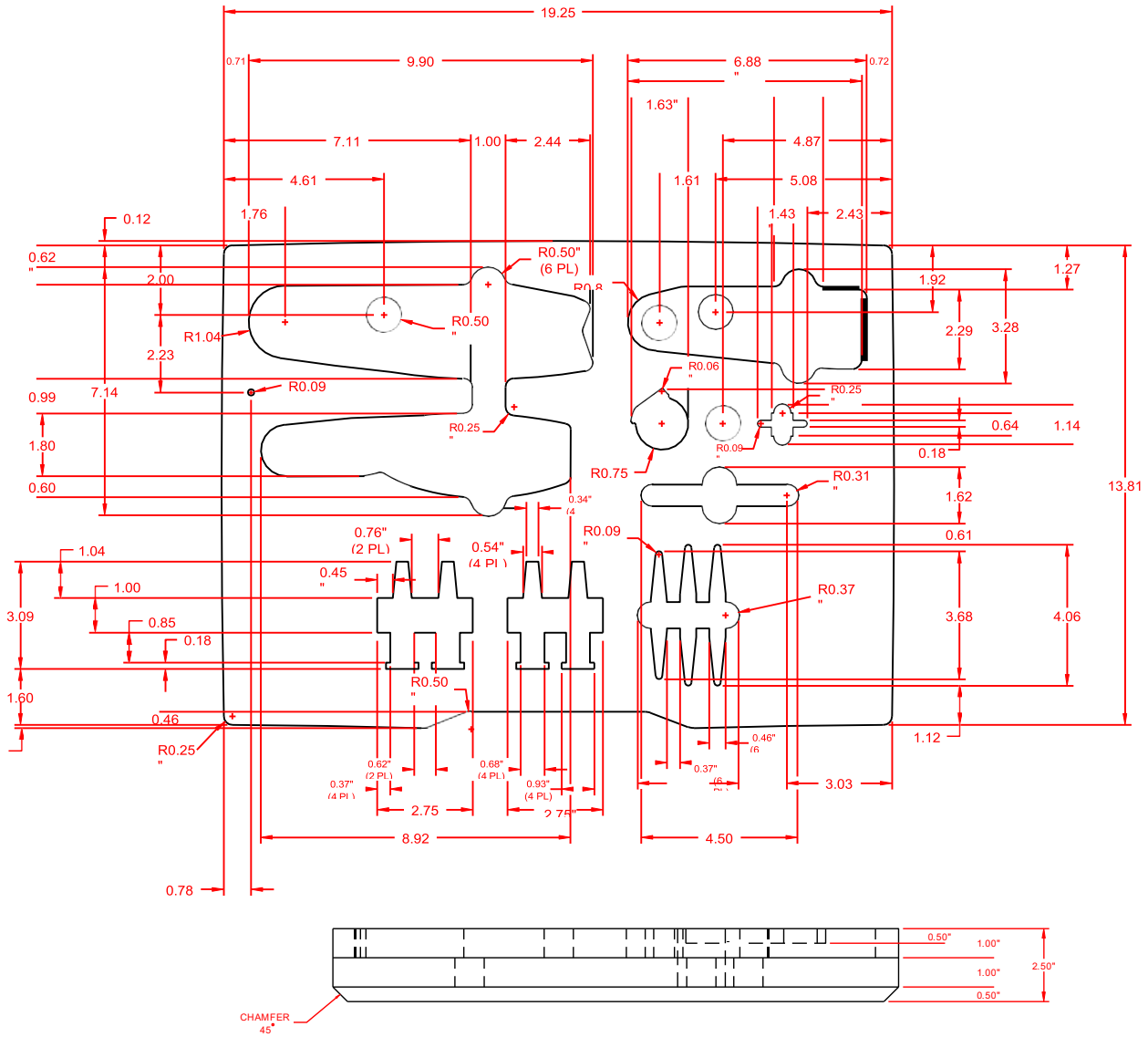


FIGURE 18 Plastic case. INSERT, BOTTOM OF CASE

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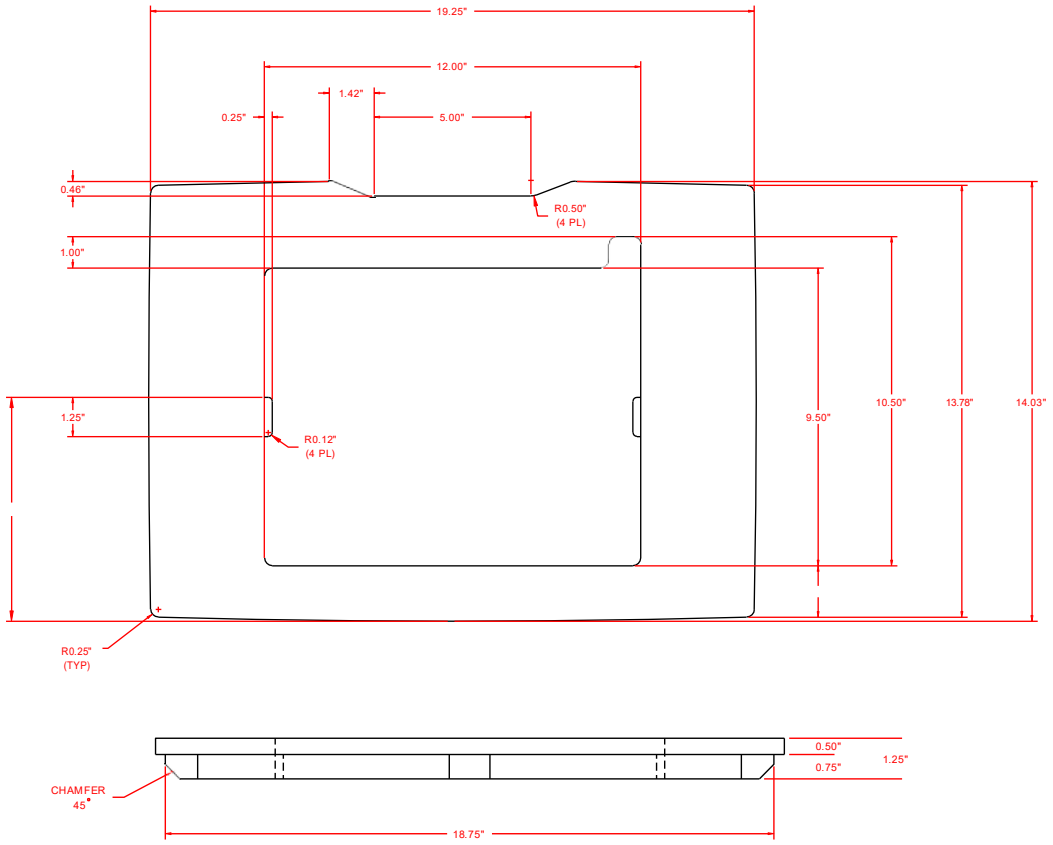


FIGURE 19 Plastic case. INSERT, TOP OF CASE

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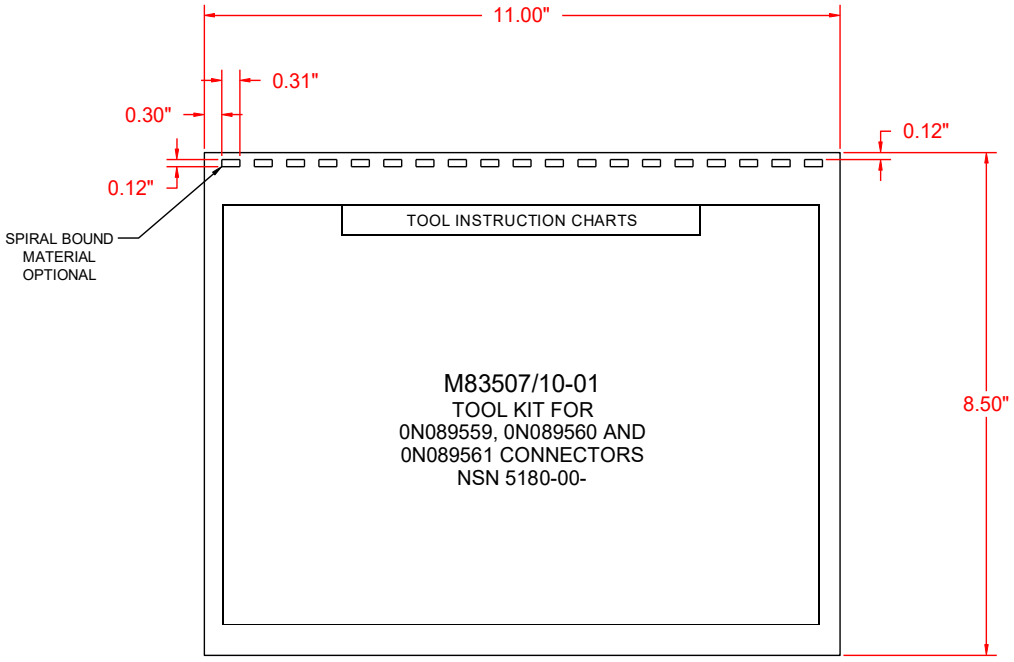


FIGURE 20 Plastic case. Instruction sheet.

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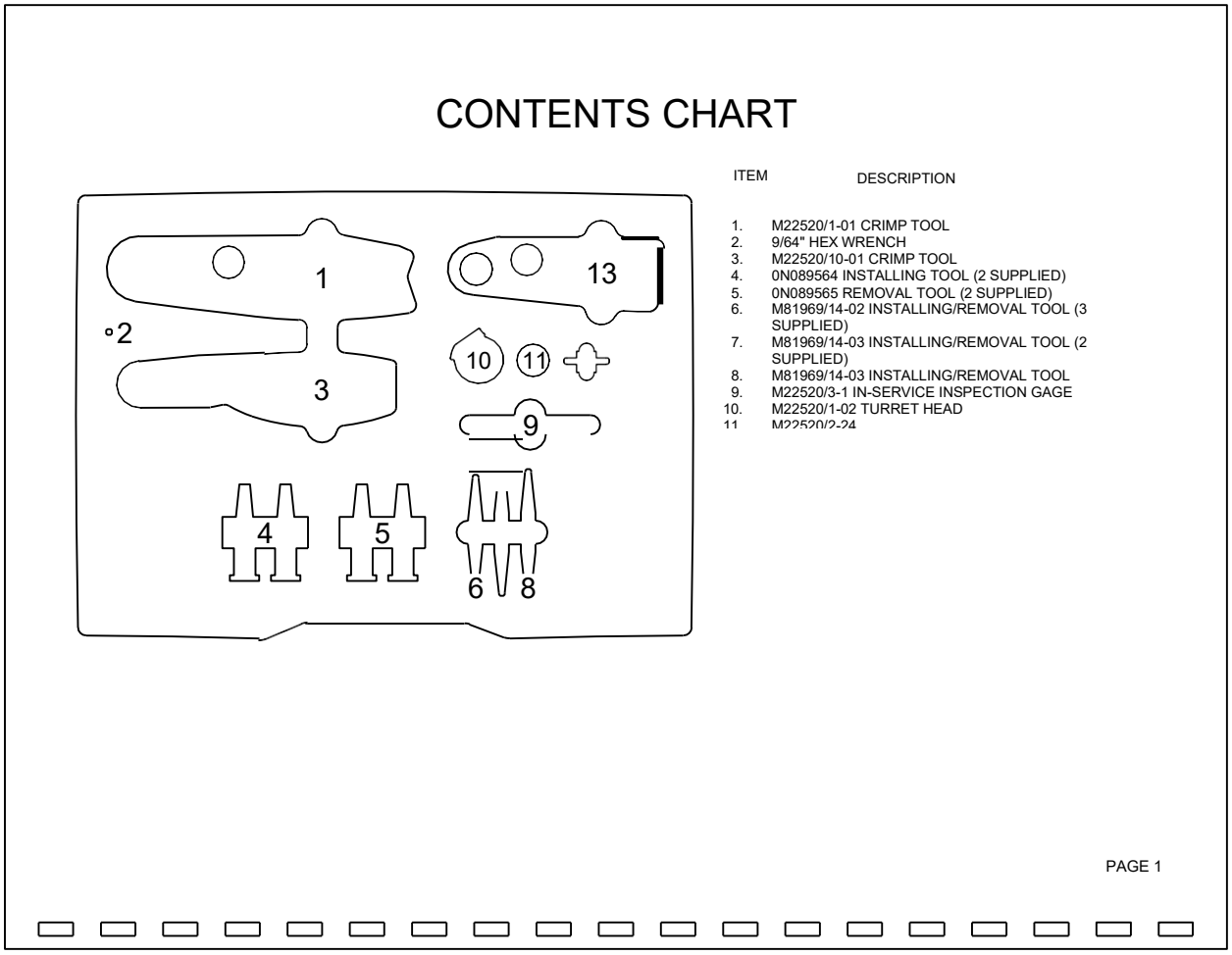
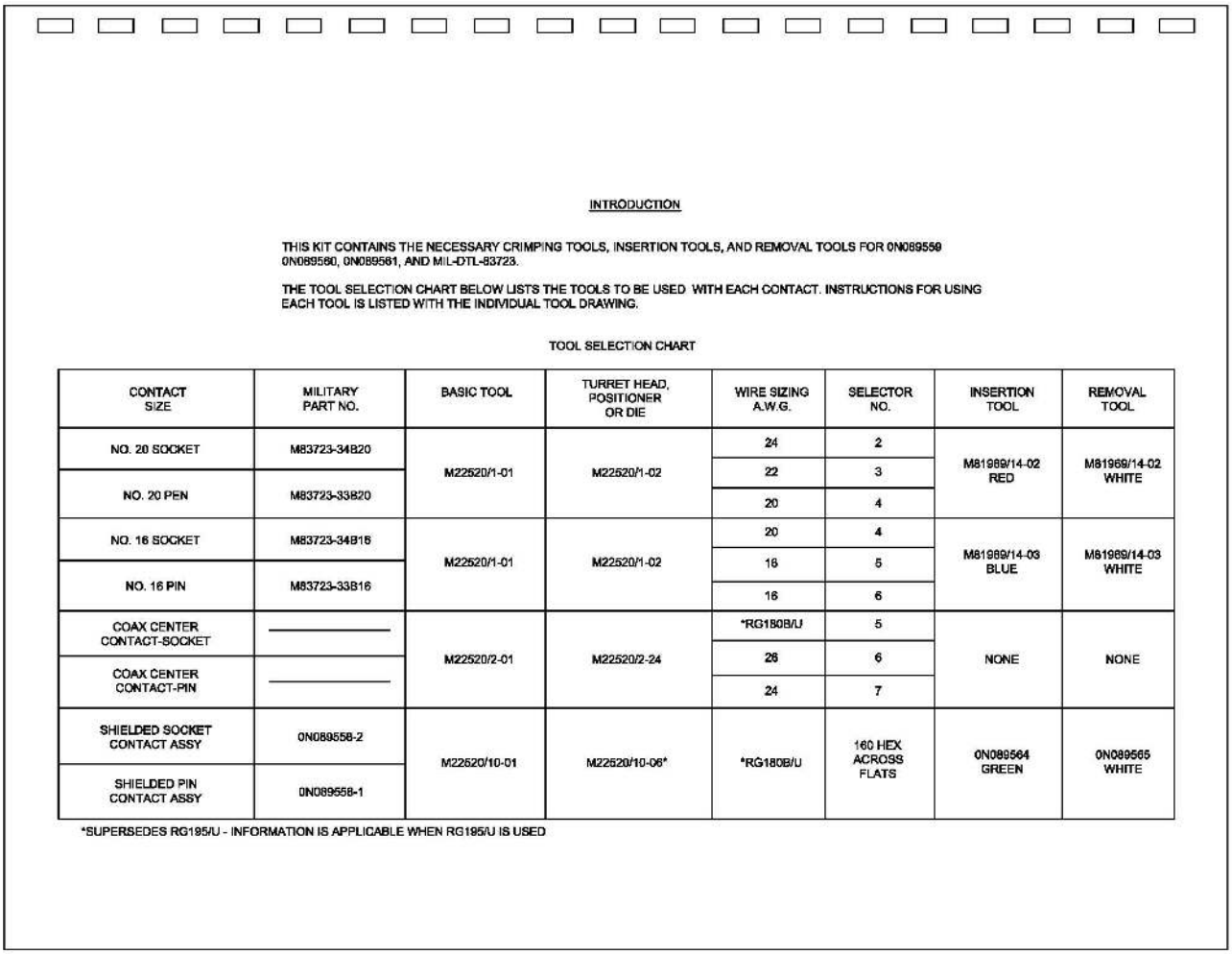


FIGURE 21 Plastic case. Instruction sheet.

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INTRODUCTION

THIS KIT CONTAINS THE NECESSARY CRIMPING TOOLS, INSERTION TOOLS, AND REMOVAL TOOLS FOR 0N089559 0N089560, 0N089561, AND MIL-DTL-83723.

THE TOOL SELECTION CHART BELOW LISTS THE TOOLS TO BE USED WITH EACH CONTACT. INSTRUCTIONS FOR USING EACH TOOL IS LISTED WITH THE INDIVIDUAL TOOL DRAWING.

TOOL SELECTION CHART

CONTACT SIZE	MILITARY PART NO.	BASIC TOOL	TURRET HEAD, POSITIONER OR DIE	WIRE SIZING A.W.G.	SELECTOR NO.	INSERTION TOOL	REMOVAL TOOL
NO. 20 SOCKET	M83723-34B20	M22520/1-01	M22520/1-02	24	2	M81969/14-02 RED	M81969/14-02 WHITE
				22	3		
NO. 20 PEN	M83723-33B20			20	4		
NO. 16 SOCKET	M83723-34B16	M22520/1-01	M22520/1-02	20	4	M81969/14-03 BLUE	M81969/14-03 WHITE
				18	5		
NO. 16 PIN	M83723-33B16			16	6		
COAX CENTER CONTACT-SOCKET	———	M22520/2-01	M22520/2-24	*RG180B/U	5	NONE	NONE
				26	6		
COAX CENTER CONTACT-PIN	———			24	7		
SHIELDED SOCKET CONTACT ASSY	0N089558-2	M22520/10-01	M22520/10-06*	*RG180B/U	160 HEX ACROSS FLATS	0N089564 GREEN	0N089565 WHITE
SHIELDED PIN CONTACT ASSY	0N089558-1						

*SUPERSEDES RG195/U - INFORMATION IS APPLICABLE WHEN RG195/U IS USED

FIGURE 22 Plastic case. Instruction sheet.

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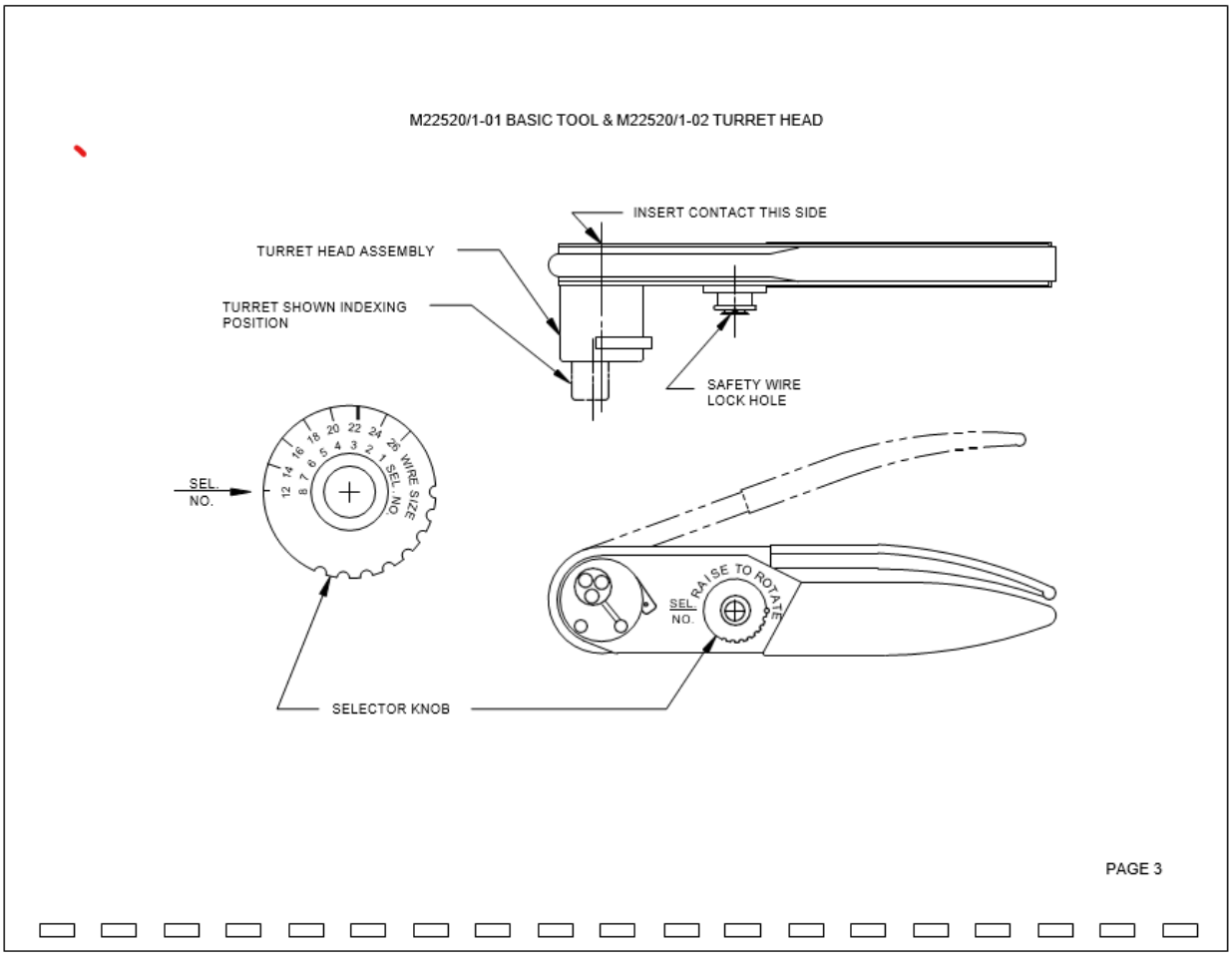


FIGURE 23 Plastic case. Instruction sheet.

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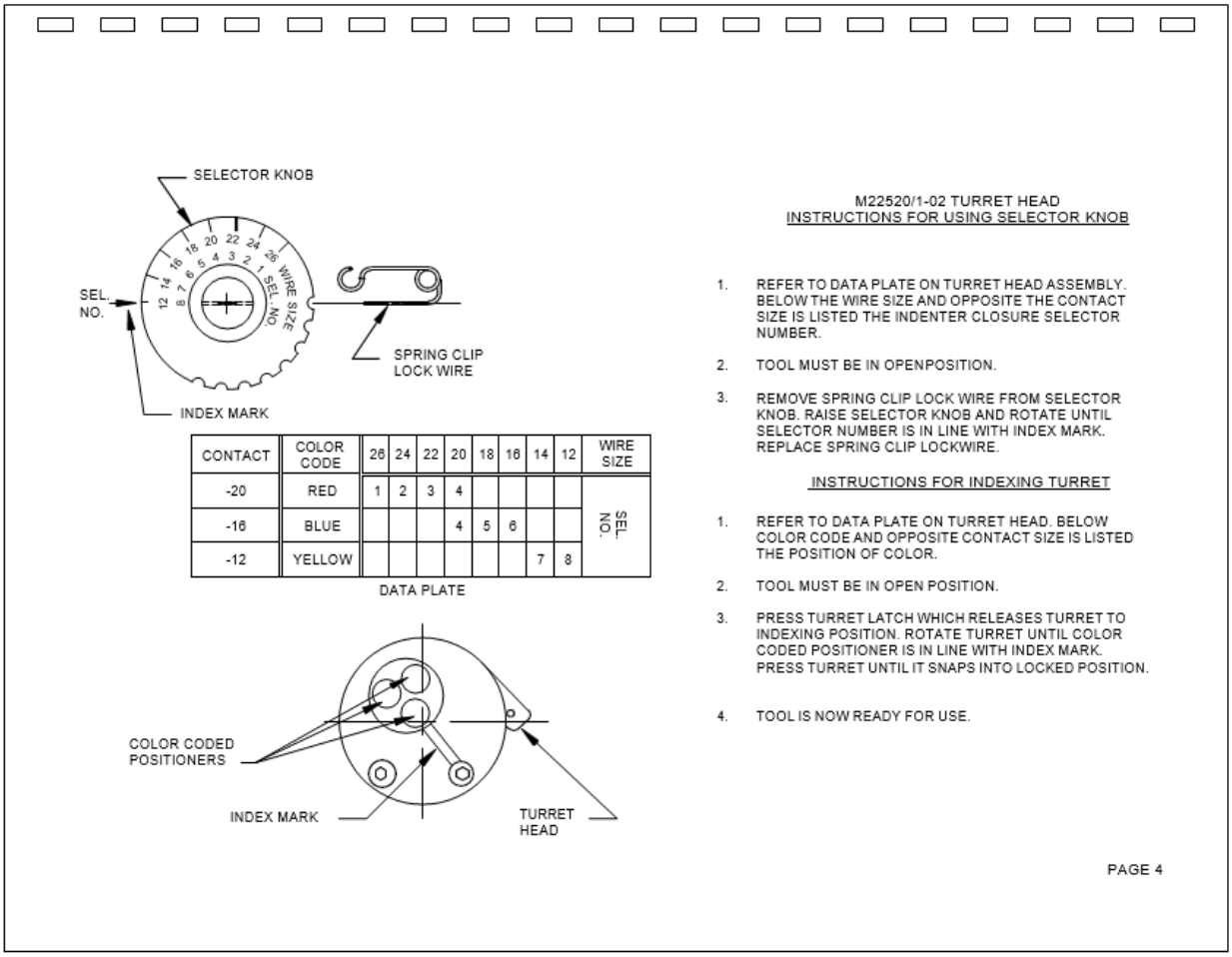


FIGURE 24 Plastic case. Instruction sheet.

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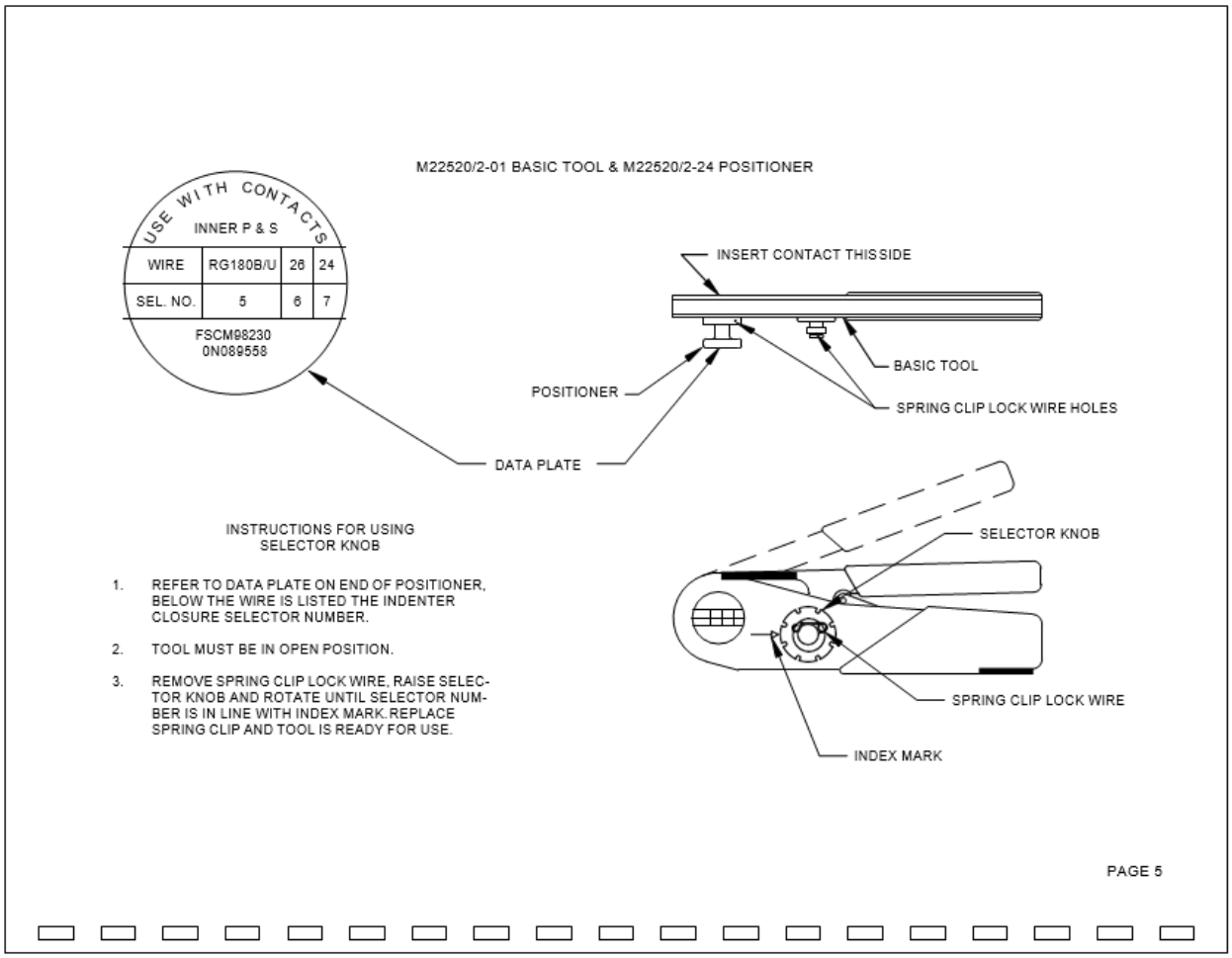


FIGURE 25 Plastic case. Instruction sheet.

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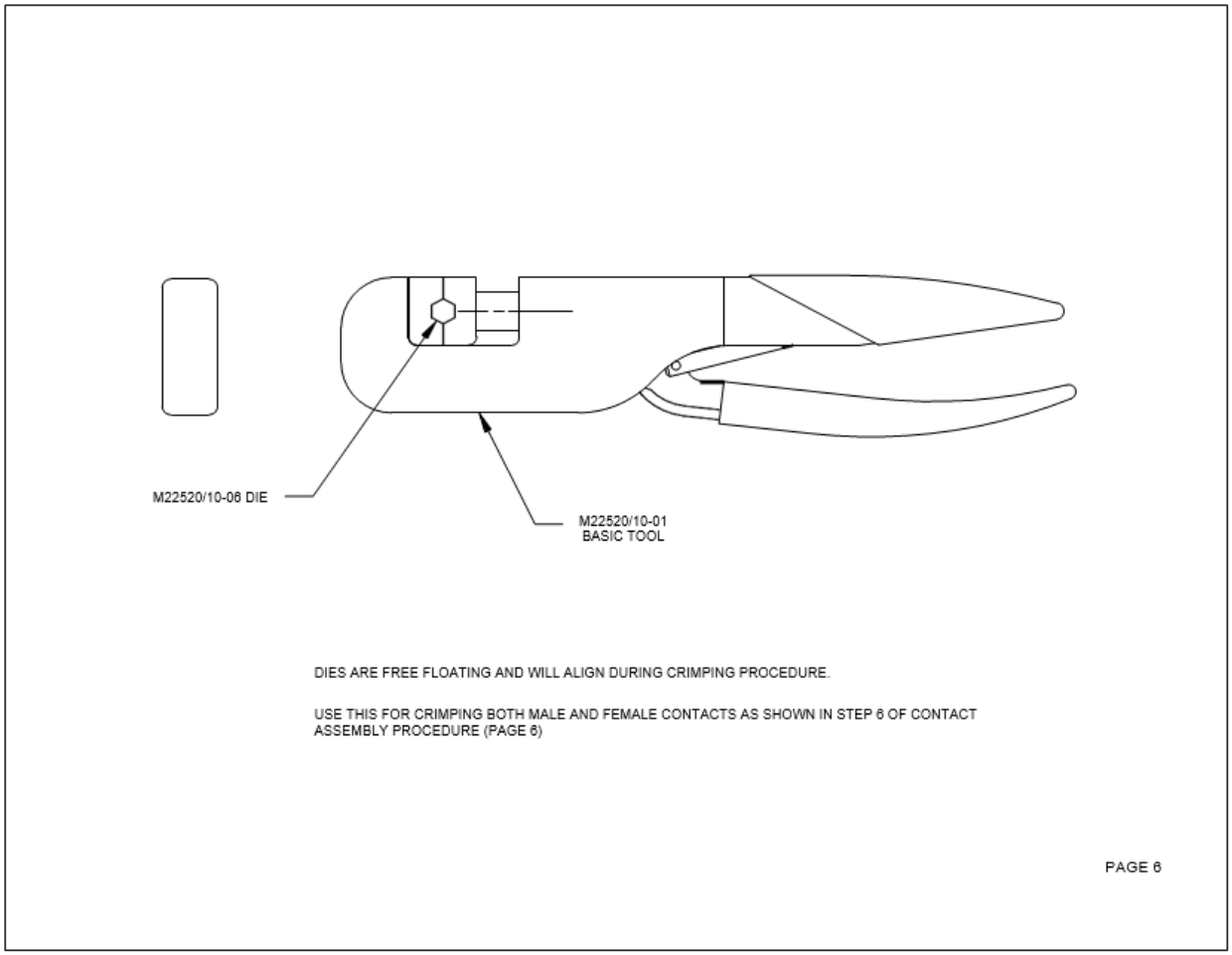


FIGURE 26 Plastic case. Instruction sheet.

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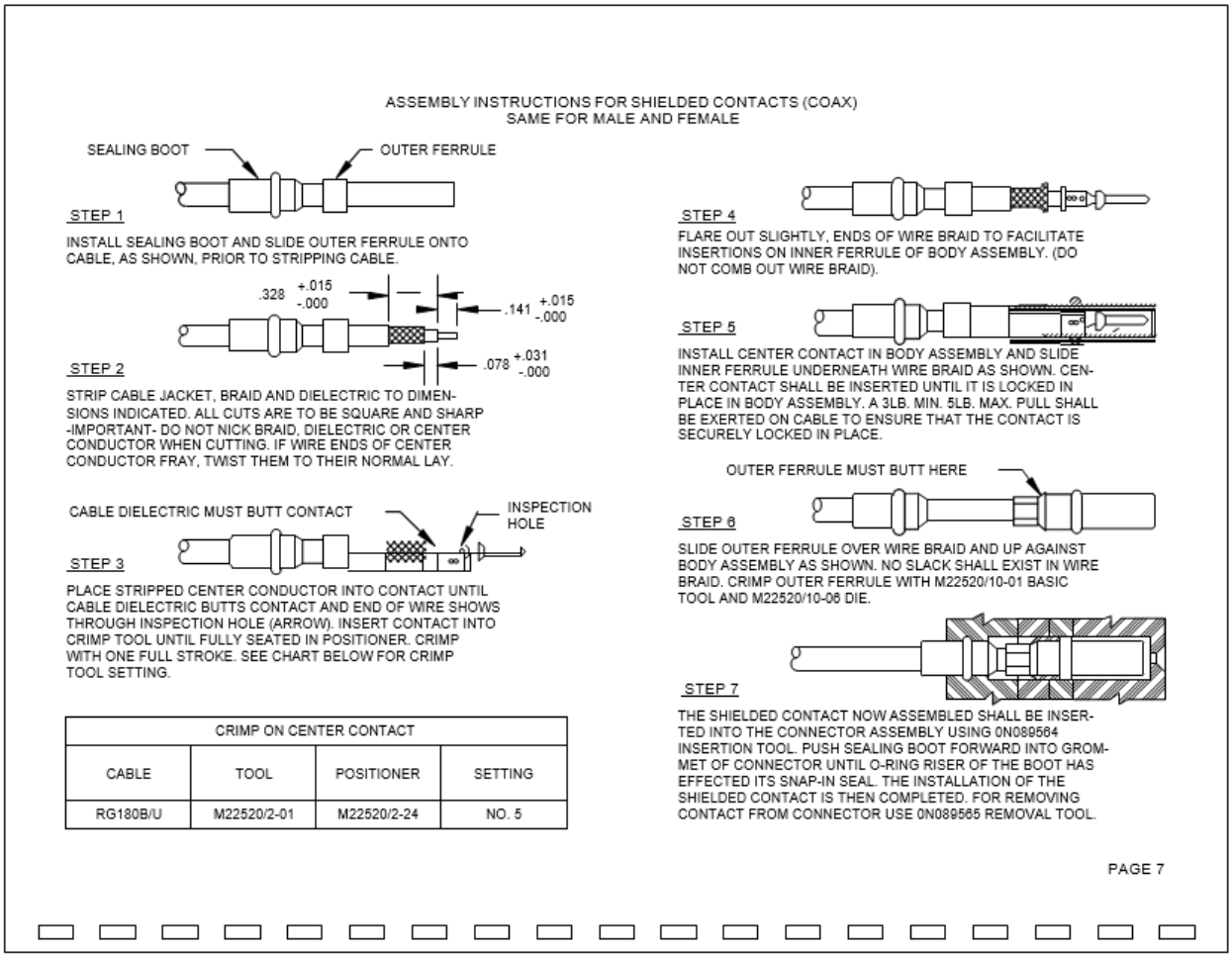


FIGURE 27 Plastic case. Instruction sheet.

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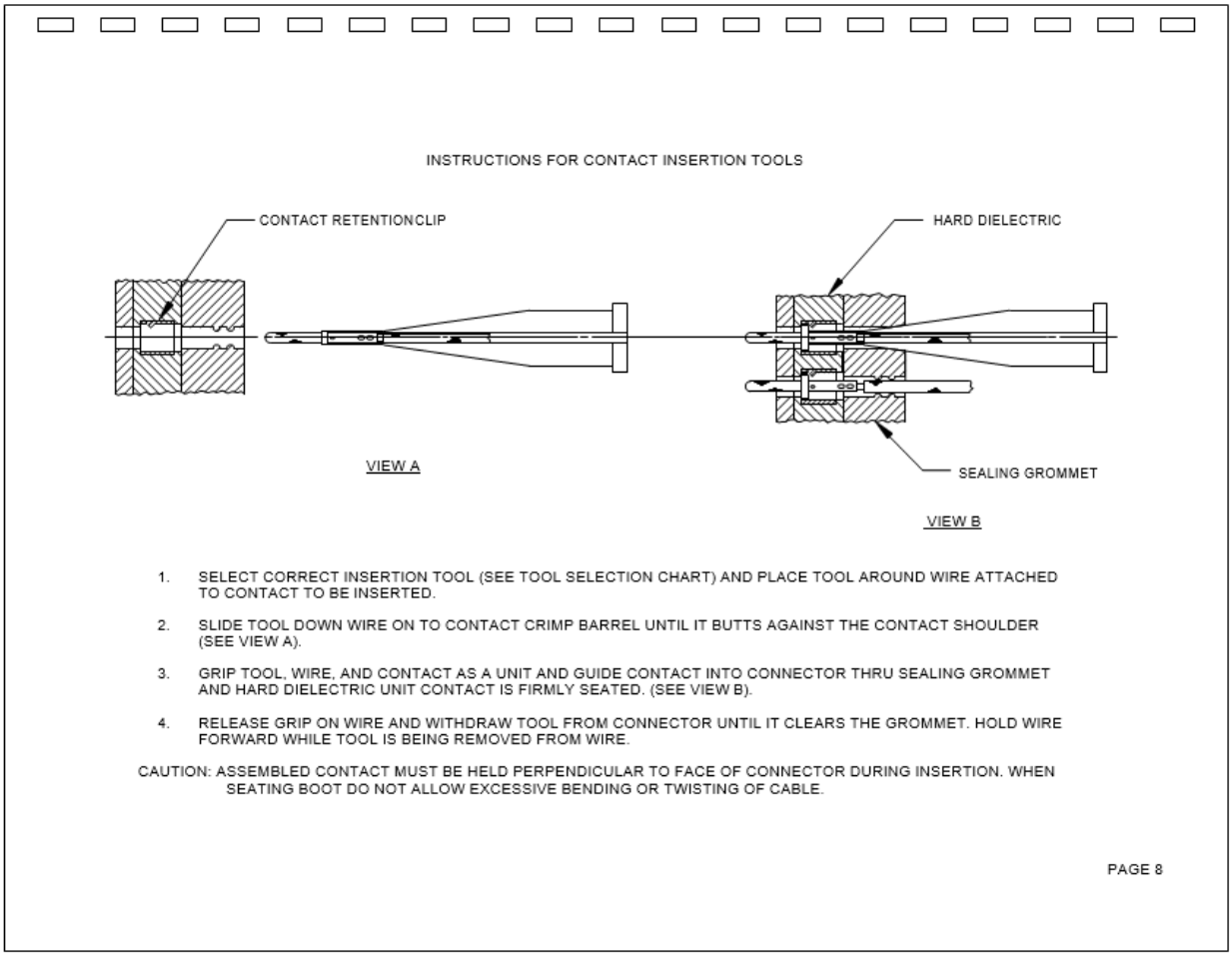


FIGURE 28 Plastic case. Instruction sheet.

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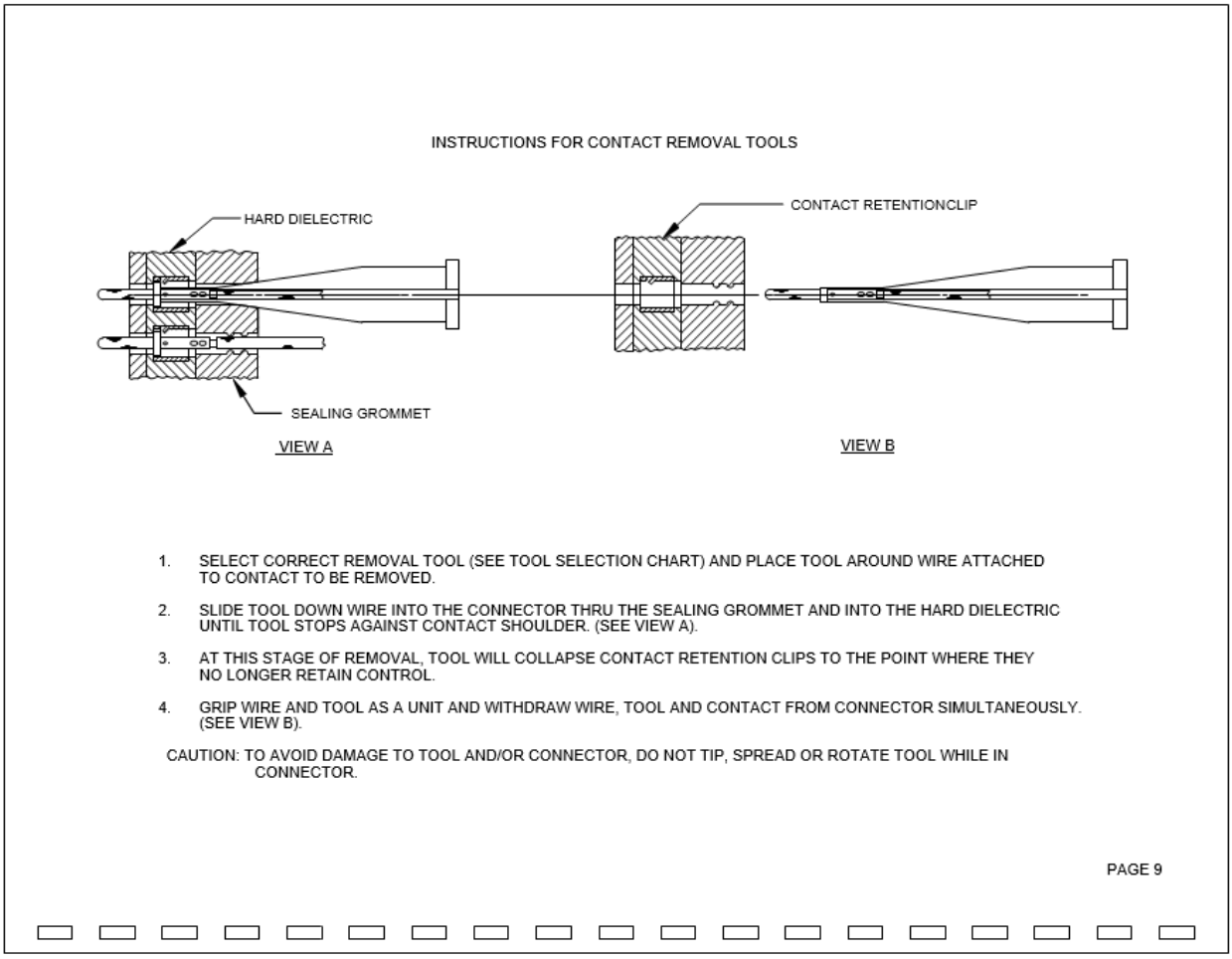


FIGURE 29 Plastic case. Instruction sheet.

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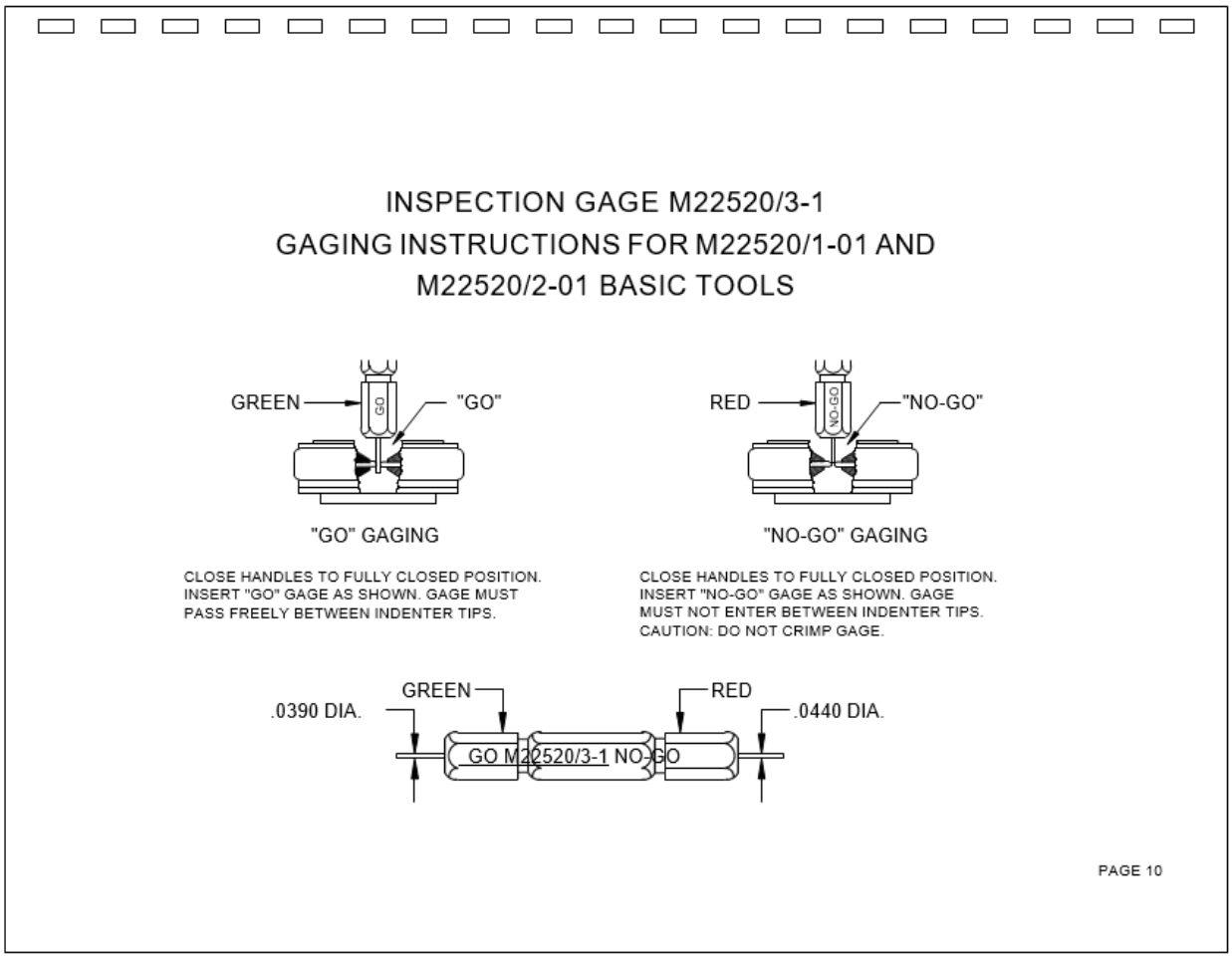


FIGURE 30 Plastic case. Instruction sheet.

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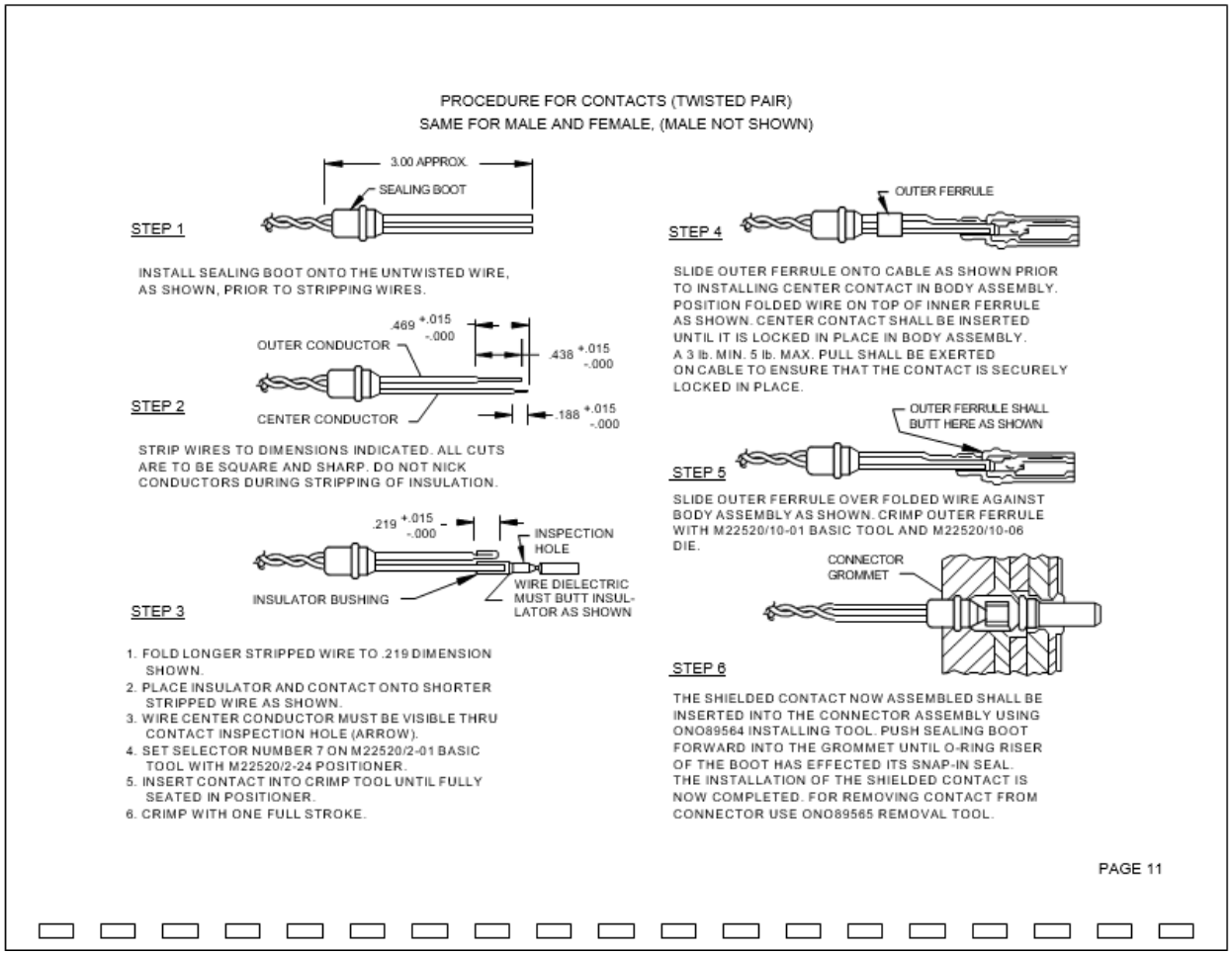


FIGURE 31 Plastic case. Instruction sheet.

REQUIREMENTS:

Material:

Metal case and cover (metal, category 3):Sheet metal, 21 gage in accordance with ASTM-A794/A794M, gray baked enamel on all metal parts.

Plastic case: Pelican

Instruction sheets and contents charts: White vinyl 0.020 inch (0.51 mm) thick. Insert and retainer pad: Fire retardant foamed polyethylene in accordance with A-A-59135 or A-A-59136; color is optional.

Binder (metal case): Looseleaf, three ring, 0.562 inch (14.27 mm) +0,062 inch (1.57 mm) I.D., 2.75 inches (69.9 mm) center distance between rings, securely riveted to inside of cover as shown.

Clasps, draw pull type with provision for locking device, handles, and hinges: Plated to resist corrosion,

Nameplate: In accordance with MIL-DTL-19834,

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Instruction sheets and contents chart: Lettering shall be legible and protected with a transparent coating. Contents chart (centered on inside of case cover) shall have pressure sensitive adhesive backing and show the insert cavity configurations and the respective tool part numbers; a list of NSN's will also be supplied.

Part numbers:

M83507/10-01 Tool kit for ON089559, ON089560, and ON089561 connectors.

M83507/10-02 Tool case assembly

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83507, this document references the following:

A-A-59135
A-A-59136
ASTM-A794/A794M
MIL-DTL-19834
MIL-DTL-83507
MIL-DTL-83723
SAE-AS22520/1
SAE-AS22520/2
SAE-AS22520/3
SAE-AS22520/10
SAE-AS81969/14

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy – AS
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army - MI, AR, AV, WC, GL
Navy - SH, EC, OS
Air Force – 17

(Project 5180-2025-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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